5

Product Measures, Convolutions, and Transforms

5.1 Product spaces and product measures

Given two measure spaces $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$, is it possible to construct a measure $\mu \equiv \mu_1 \times \mu_2$ on a σ-algebra on the product space $\Omega_1 \times \Omega_2$ such that $\mu(A \times B) = \mu_1(A)\mu_2(B)$ for $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$? This section is devoted to studying this question.

Definition 5.1.1: Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be measurable spaces.

(a) $\Omega_1 \times \Omega_2 \equiv \{(\omega_1, \omega_2) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\}$, the set of all ordered pairs, is called the *(Cartesian) product of* Ω_1 *and* Ω_2.

(b) The set $A_1 \times A_2$ with $A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2$ is called a *measurable rectangle*. The collection of measurable rectangles will be denoted by $\mathcal{C}$.

(c) The *product σ-algebra of* $\mathcal{F}_1$ *and* $\mathcal{F}_2$ on $\Omega_1 \times \Omega_2$, denoted by $\mathcal{F}_1, \times \mathcal{F}_2$, is the smallest σ-algebra generated by $\mathcal{C}$, i.e.,

$$\mathcal{F}_1 \times \mathcal{F}_2 \equiv \sigma\langle\{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\}\rangle.$$

(d) $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$ is called the *product measurable space.*

Starting with the definition of μ on the class $\mathcal{C}$ of measurable rectangles by

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \tag{1.1}$$

for all $A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2$, one can extend it to a measure on the algebra $\mathcal{A}$ of all finite unions of disjoint measurable rectangles in a natural way. Indeed, the extension to $\mathcal{A}$ is obtained simply by assigning the μ-measure of a finite union of disjoint measurable rectangles as the sum of the μ-measures of the corresponding individual measurable rectangles. Then, by the extension theorem (cf. Theorem 1.3.3), it can be further extended to a complete measure μ on a σ-algebra containing $\mathcal{F}_1 \times \mathcal{F}_2$, defined in (c) above. However, to evaluate $\mu(A)$ for an arbitrary A in $\mathcal{F}_1 \times \mathcal{F}_2$ that is not a measurable rectangle and to evaluate $\int h d\mu$ for arbitrary measurable functions h, some further work is needed. For the case of the product of two measure spaces, here an alternate approach that yields a direct way of computing these quantities is presented. The extension theorem approach will be used for the more general case of products of finitely many measure spaces in Section 5.3. The case of products of infinitely many probability spaces will be discussed in Chapter 6.

Definition 5.1.2:

(a) Let $A \in \mathcal{F}_1 \times \mathcal{F}_2$. Then, for any $\omega_1 \in \Omega_1$, the set

$$A_{1\omega_1} \equiv \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in A\} \tag{1.2}$$

is called the ω_1-*section of* A and for any $\omega_2 \in \Omega_2$, the set $A_{2\omega_2} \equiv \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in A\}$ is called the ω_2-*section* of A.

(b) If $f : (\Omega_1 \times \Omega_2) \to \Omega_3$ is a $\langle \mathcal{F}_1 \times \mathcal{F}_2, \mathcal{F}_3 \rangle$ measurable mapping from $\Omega_1 \times \Omega_2$ into some measurable space $(\Omega_3, \mathcal{F}_3)$, then the ω_1-*section of* f is the function $f_{1\omega_1} : \Omega_2 \to \Omega_3$, given by

$$f_{1\omega_1}(\omega_2) = f(\omega_1, \omega_2),\ \omega_2 \in \Omega_2. \tag{1.3}$$

Similarly, one may define the ω_2-*sections* of f by $f_{2\omega_2}(\omega_1) = f(\omega_1, \omega_2)$, $\omega_1 \in \Omega_1$. The following result shows that the ω_1-sections of a $\mathcal{F}_1 \times \mathcal{F}_2$-measurable function is $\mathcal{F}_2$-measurable when considered as a function of $\omega_2 \in \Omega_2$.

Proposition 5.1.1: *Let* $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$ *be a product space,* $A \in \mathcal{F}_1 \times \mathcal{F}_2$ *and let* $f : \Omega_1 \times \Omega_2 \to \Omega_3$ *be a* $\langle \mathcal{F}_1 \times \mathcal{F}_2, \mathcal{F}_3 \rangle$*-measurable function.*

(i) For every $\omega_1 \in \Omega_1$, $A_{1\omega_1} \in \mathcal{F}_2$ *and for every* $\omega_2 \in \Omega_2$, $A_{2\omega_2} \in \mathcal{F}_1$.

(ii) For every $\omega_1 \in \Omega_1$, $f_{1\omega_1}$ *is* $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$*-measurable and for every* $\omega_2 \in \Omega_2$, $f_{2\omega_2}$ *is* $\langle \mathcal{F}_1, \mathcal{F}_3 \rangle$*-measurable.*

Proof: Fix $\omega_1 \in \Omega_1$. Define the function $g : \Omega_2 \to \Omega_1 \times \Omega_2$ by

$$g(\omega_2) = (\omega_1, \omega_2),\ \omega_2 \in \Omega_2.$$

Note that for any measurable rectangle $A \equiv A_1 \times A_2 \in \mathcal{F}_1 \times \mathcal{F}_2$,

$$A_{1\omega_1} = \begin{cases} A_2 & \text{if} \quad \omega_1 \in A_1 \\ \emptyset & \text{if} \quad \omega_1 \notin A_1 \end{cases}$$

and hence $g^{-1}(A_1 \times A_2) \in \mathcal{F}_2$. Since the class of all measurable rectangles generates $\mathcal{F}_1 \times \mathcal{F}_2$, for fixed ω_1 in Ω_1, g is $\langle \mathcal{F}_2, \mathcal{F}_1 \times \mathcal{F}_2 \rangle$-measurable. Therefore, for A in $\mathcal{F}_1 \times \mathcal{F}_2$, $A_{1\omega_1} = g^{-1}(A) \in \mathcal{F}_2$ and for f as given, $f_{1\omega_1} = f \circ g$ is $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$-measurable. This proves (i) and (ii) for the ω_1-sections. The proof for the ω_2-sections are similar. $\square$

Now suppose μ_1 and μ_2 are measures on $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, respectively. Then, for any set $A \in \mathcal{F}_1 \times \mathcal{F}_2$, for all $\omega_1 \in \Omega_1$, $A_{1\omega_1} \in \mathcal{F}_2$ and hence $\mu_2(A_{1\omega_1})$ is well defined. If this were an $\mathcal{F}_1$-measurable function, then one might define a set function on $\mathcal{F}_1 \times \mathcal{F}_2$ by

$$\mu_{12}(A) = \int_{\Omega_1} \mu_2(A_{1\omega_1}) \mu_1(d\omega_1). \tag{1.4}$$

And similarly, reversing the order of μ_1 and μ_2, one might define a second set function

$$\mu_{21}(A) = \int_{\Omega_2} \mu_1(A_{2\omega_2}) \mu_2(d\omega_2), \tag{1.5}$$

provided that $\mu_1(A_{2\omega_2})$ is $\mathcal{F}_2$-measurable. Note that for the measurable rectangles $A = A_1 \times A_2$, $\mu_{12}(A) = \mu_1(A_1)\mu_2(A_2) = \mu_{21}(A)$ and thus both μ_{12} and μ_{21} coincide (with the *product* measure μ) on the class $\mathcal{C}$ of all measurable rectangles. This implies that if the product measure μ is *unique* on $\mathcal{F}_1 \times \mathcal{F}_2$, and μ_{12} and μ_{21} are measures on $\mathcal{F}_1 \times \mathcal{F}_2$, then μ_{12}, μ_{21} and μ must coincide on the σ-algebra $\mathcal{F}_1 \times \mathcal{F}_2$. Then, one can evaluate $\mu(A)$ for any set $A \in \mathcal{F}_1 \times \mathcal{F}_2$ using either of the relations (1.4) or (1.5). The following result makes this heuristic discussion rigorous.

Theorem 5.1.2: *Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$ be σ-finite measure spaces. Then,*

(i) for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$, the functions $\mu_2(A_{1\omega_1})$ and $\mu_1(A_{2\omega_2})$ are $\mathcal{F}_1$- and $\mathcal{F}_2$-measurable, respectively.

(ii) The set functions μ_{12} and μ_{21}, given by (1.4) and (1.5) respectively, are measures on $\mathcal{F}_1 \times \mathcal{F}_2$, satisfying $\mu_{12}(A) = \mu_{21}(A)$ for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$.

(iii) Further, $\mu_{12} = \mu_{21} \equiv \mu$ is σ-finite and it is the only measure satisfying

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \quad \text{for all} \quad A_1 \times A_2 \in \mathcal{C}.$$

Proof: First suppose that μ_1 and μ_2 are finite measures. Define

$$\mathcal{L} = \big\{ A \in \mathcal{F}_1 \times \mathcal{F}_2 : \mu_2(A_{1\omega_1}) \quad \text{is a} \quad \langle \mathcal{F}_1, \mathcal{B}(\mathbb{R}) \rangle\text{-measurable function} \big\}.$$

For $A = \Omega_1 \times \Omega_2$, $\mu_2(A_{1\omega_1}) \equiv \mu_2(\Omega_2)$ for all $\omega_1 \in \Omega_1$ and hence $\Omega_1 \times \Omega_2 \in \mathcal{L}$. Next, let $A, B \in \mathcal{L}$ with $A \subset B$. Then, it is easy to check that $(A \setminus B)_{1\omega_1} = A_{1\omega_1} \setminus B_{1\omega_1}$. Since μ_2 is a finite measure and $A, B \in \mathcal{L}$, it follows that $\mu_2((A \setminus B)_{1\omega_1}) = \mu_2(A_{1\omega_1} \setminus B_{1\omega_1}) = \mu_2(A_{1\omega_1}) - \mu_2(B_{1\omega_1})$ is $\langle \mathcal{F}_1, \mathcal{B}(\mathbb{R})\rangle$-measurable. Thus, $A \setminus B \in \mathcal{L}$. Finally, let $\{B_n\}_{n\geq 1} \subset \mathcal{L}$ be such that $B_n \subset B_{n+1}$ for all $n \geq 1$. Then for any $\omega_1 \in \Omega_1$, $(B_n)_{1\omega_1} \subset (B_{n+1})_{1\omega_1}$ for all $n \geq 1$. Hence by the MCT,

$$\infty > \mu_2\Big(\Big(\bigcup_{n\geq 1} B_n\Big)_{1\omega_1}\Big) = \mu_2\Big(\bigcup_{n\geq 1} (B_n)_{1\omega_1}\Big) = \lim_{n\to\infty} \mu_2\big((B_n)_{1\omega_1}\big)$$

for all $\omega_1 \in \Omega_1$. By Proposition 2.1.5, this implies that $\mu_2\big((\bigcup_{n\geq 1} B_n)_{1\omega_1}\big)$ is $\langle \mathcal{F}_1, \mathcal{B}(\mathbb{R})\rangle$-measurable, and hence $\bigcup_{n\geq 1} B_n \in \mathcal{L}$. Thus, $\mathcal{L}$ is a λ-system. For $A = A_1 \times A_2 \in \mathcal{C}$, $\mu_2(A_{1\omega_1}) = \mu_2(A_2) I_{A_1}(\omega_1)$ and hence $\mathcal{C} \subset \mathcal{L}$. Since $\mathcal{C}$ is a π-system, by Corollary 1.1.3, it follows that $\mathcal{L} = \mathcal{F}_1 \times \mathcal{F}_2$. Thus, $\mu_2(A_{1\omega_1})$, considered as a function of ω_1, is $\langle \mathcal{F}_1, \mathcal{B}(\mathbb{R})\rangle$-measurable for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$, proving (i).

Next, consider part (ii). By part (i), μ_{12} is a well-defined set function on $\mathcal{F}_1 \times \mathcal{F}_2$. It is easy to check that μ_{12} is a measure on $\mathcal{F}_1 \times \mathcal{F}_2$ (Problem 5.1). Similarly, μ_{21} is a well-defined measure on $\mathcal{F}_1 \times \mathcal{F}_2$. Since $\mu_{12}(A) = \mu_{21}(A) = \mu_1(A_1)\mu_2(A_2)$ for all $A = A_1 \times A_2 \in \mathcal{C}$ and $\mathcal{C}$ is a π-system generating $\mathcal{F}_1 \times \mathcal{F}_2$, it follows from Theorem 1.2.4 that $\mu_{12}(A) = \mu_{21}(A)$ for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$. This proves (ii) for the case where $\mu_i(\Omega_i) < \infty$, $i = 1, 2$.

Next, suppose that μ_i's are σ-finite. Then, there exist disjoint sets $\{B_{in}\}_{n\geq 1} \subset \mathcal{F}_i$, such that $\bigcup_{n\geq 1} B_{in} = \Omega_i$ and $\mu_i(B_{in}) < \infty$ for all $n \geq 1$, $i = 1, 2$. Define the finite measures

$$\mu_{in}(D) = \mu_i(D \cap B_{in}), \quad D \in \mathcal{F}_i,$$

for $n \geq 1$, $i = 1, 2$. The arguments above with μ_i replaced by μ_{in} imply that for any $A \in \mathcal{F}_1 \times \mathcal{F}_2$, $\mu_{2n}(A_{1\omega_1})$ is $\langle \mathcal{F}_1, \mathcal{B}(\mathbb{R})\rangle$-measurable for all $n \geq 1$. Since μ_2 is a measure on $\mathcal{F}_2$, $\mu_2(A_{1\omega_1}) = \sum_{n=1}^{\infty} \mu_{2n}(A_{\omega_1})$ and hence, considered as a function of ω_1, it is $\langle \mathcal{F}_1, \mathcal{B}(\bar{\mathbb{R}})\rangle$-measurable for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$. Thus, the set function μ_{12} of (1.4) is well defined in the σ-finite case as well. Similarly, μ_{21} of (1.5) is a well-defined set function on $\mathcal{F}_1 \times \mathcal{F}_2$. Let $\mu_{12}^{(m,n)}$ and $\mu_{21}^{(m,n)}$, respectively, denote the set functions defined by (1.4) and (1.5) with μ_1 replaced by μ_{1m} and μ_2 replaced by μ_{2n}, $m \geq 1$, $n \geq 1$. Then, by repeated use of the MCT,

$$\begin{aligned}
\mu_{12}(A) &= \int_{\Omega_1} \mu_2(A_{1\omega_1})\mu_1(d\omega_1) \\
&= \sum_{m=1}^{\infty} \Big(\int_{B_{1m}} \sum_{n=1}^{\infty} \mu_2(A_{1\omega_1} \cap B_{2n})\Big)\mu_1(d\omega_1) \\
&= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \int_{B_{1m}} \mu_2(A_{1\omega_1} \cap B_{2n})\mu_1(d\omega_1)
\end{aligned}$$

$$= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \mu_{12}^{(m,n)}(A), \quad A \in \mathcal{F}_1 \times \mathcal{F}_2 \tag{1.6}$$

and similarly,

$$\mu_{21}(A) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} \mu_{21}^{(m,n)}(A), \quad A \in \mathcal{F}_1 \times \mathcal{F}_2. \tag{1.7}$$

Since $\mu_{12}^{(m,n)}$ and $\mu_{21}^{(m,n)}$ are (finite) measures, it is easy to check that μ_{12} and μ_{21} are measures on $\mathcal{F}_1 \times \mathcal{F}_2$. Also, by the finite case, $\mu_{12}^{(m,n)}(A_1 \times A_2) = \mu_{21}^{(m,n)}(A_1 \times A_2)$ for all $n \geq 1$, $m \geq 1$ and hence

$$\mu_{12}(A_1 \times A_2) = \mu_{21}(A_1 \times A_2) \quad \text{for all} \quad A_1 \times A_2 \in \mathcal{C}.$$

Next note that $\{B_{1m} \times B_{2n} : m \geq 1, n \geq 1\}$ is a partition of $\Omega_1 \times \Omega_2$ by $\mathcal{F}_1 \times \mathcal{F}_2$ sets and by (1.6) and (1.7), for all $m \geq 1$, $n \geq 1$,

$$\mu_{12}(B_{1m} \times B_{2n}) = \mu_1(B_{1m})\mu_2(B_{2n}) = \mu_{21}(B_{1m} \times B_{2n}) < \infty.$$

Hence, μ_{12} and μ_{21} are σ-finite on $\mathcal{F}_1 \times \mathcal{F}_2$. Since μ_{12} and μ_{21} agree on $\mathcal{C}$ and $\mathcal{C}$ is a π-system generating the product σ-algebra, it follows that $\mu_{12}(A) = \mu_{21}(A)$ for all $A \in \mathcal{F}_1 \times \mathcal{F}_2$ and it is the unique measure satisfying $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ for all $A_1 \times A_2 \in \mathcal{C}$. This completes the proof of the theorem. □

Remark 5.1.1: In the above theorem, the σ-finiteness condition on the measures μ_1 and μ_2 cannot be dropped. For example, let $\Omega_1 = \Omega_2 = [0,1]$, $\mathcal{F}_1 = \mathcal{F}_2 = \mathcal{B}([0,1])$, $\mu_1 =$ the Lebesgue measure and $\mu_2 =$ the counting measure. Clearly μ_2 is not σ-finite since $[0,1]$ is uncountable. Let A be the diagonal set in the product space $[0,1]\times[0,1]$, i.e., $A = \{(\omega_1,\omega_2) \in \Omega_1 \times \Omega_2 : \omega_1 = \omega_2\}$. Then $A \in \mathcal{F}_1 \times \mathcal{F}_2$, $A_{1\omega_1} = \{\omega_1\}$ and $A_{2\omega_2} = \{\omega_2\}$. Further, $\mu_2(A_{1\omega_1}) = 1$ for all ω_1 and $\mu_1(A_{2\omega_2}) = 0$ for all ω_2. Thus $\mu_{12}(A) = \int_{\Omega_1} \mu_2(A_{1\omega_1})\mu_1(d\omega_1) = 1$, but $\mu_{21}(A) = \int_{\Omega_2} \mu_1(A_{2\omega_2})\mu_2(d\omega_2) = 0$, and hence, $\mu_{12}(A) \neq \mu_{21}(A)$.

Remark 5.1.2: Although this approach allows one to compute the product measure $\mu_1 \times \mu_2$ of a set $A \in \mathcal{F}_1 \times \mathcal{F}_2$, the measure space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, \mu_1 \times \mu_2)$ may not be complete even if both $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1,2$ are complete (Problem 5.2). However, the approach based on the extension theorem yields a product measure space that is complete. See Remark 5.2.2 for further discussion on the topic.

Definition 5.1.3:

(a) The unique measure μ on $\mathcal{F}_1 \times \mathcal{F}_2$ defined in Theorem 5.1.2 (iii) is called the *product measure* and is denoted by $\mu_1 \times \mu_2$.

(b) The measure space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, \mu_1 \times \mu_2)$ is called the *product measure space.*

Formulas (1.4) and (1.5) give two different ways of evaluating $\mu_1 \times \mu_2(A)$ for an $A \in \mathcal{F}_1 \times \mathcal{F}_2$. In the next section, integration of a measurable function $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ w.r.t. the product measure $\mu_1 \times \mu_2$ is considered and the above approach is extended to justify evaluation of the integral iteratively.

5.2 Fubini-Tonelli theorems

Let $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ be a $\langle \mathcal{F}_1 \times \mathcal{F}_2, \mathcal{B}(\mathbb{R})\rangle$-measurable function. Relations (1.4) and (1.5) suggest that the integral of f w.r.t. $\mu_1 \times \mu_2$ may be evaluated as iterated integrals, using the formulas

$$\int_{\Omega_1 \times \Omega_2} f(\omega_1, \omega_2)\mu_1 \times \mu_2(d(\omega_1, \omega_2)) = \int_{\Omega_2} \left[\int_{\Omega_1} f(\omega_1, \omega_2)\mu_1(d\omega_1)\right] \mu_2(d\omega_2) \tag{2.1}$$

and

$$\int_{\Omega_1 \times \Omega_2} f(\omega_1, \omega_2)\mu_1 \times \mu_2(d(\omega_1, \omega_2)) = \int_{\Omega_1} \left[\int_{\Omega_2} f(\omega_1, \omega_2)\mu_2(d\omega_2)\right] \mu_1(d\omega_1). \tag{2.2}$$

Here, the left sides of both (2.1) and (2.2) are simply the integral of f on the space $\Omega = \Omega_1 \times \Omega_2$ w.r.t. the measure $\mu = \mu_1 \times \mu_2$. The expressions on the right sides of (2.1) and (2.2) are, however, *iterated integrals*, where integrals of sections of f are evaluated first and then the resulting *sectional integrals* are integrated again to get the final expression. Conditions for the validity of (2.1) and (2.2) are provided by the Fubini-Tonelli theorems stated below.

Theorem 5.2.1: (*Tonelli's theorem*). *Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$ be σ-finite measure spaces and let $f : \Omega_1 \times \Omega_2 \to \mathbb{R}_+$ be a nonnegative $\mathcal{F}_1 \times \mathcal{F}_2$-measurable function. Then*

$$g_1(\omega_1) \equiv \int_{\Omega_2} f(\omega_1, \omega_2)\mu_2(d\omega_2) : \Omega_1 \to \bar{\mathbb{R}} \quad \textit{is} \quad \langle \mathcal{F}_1, \mathcal{B}(\bar{\mathbb{R}})\rangle\textit{-measurable} \tag{2.3}$$

and

$$g_2(\omega_2) \equiv \int_{\Omega_1} f(\omega_1, \omega_2)\mu_1(d\omega_1) : \Omega_2 \to \bar{\mathbb{R}} \quad \textit{is} \quad \langle \mathcal{F}_2, \mathcal{B}(\bar{\mathbb{R}})\rangle\textit{-measurable.} \tag{2.4}$$

Further

$$\int_{\Omega_1 \times \Omega_2} f d\mu = \int_{\Omega_1} g_1 d\mu_1 = \int_{\Omega_2} g_2 d\mu_2, \tag{2.5}$$

where $\mu = \mu_1 \times \mu_2$.

Proof: If $f = I_A$ for some A in $\mathcal{F}_1 \times \mathcal{F}_2$, the result follows from Theorem 5.1.2. By the linearity of integrals, the result now holds for all simple nonnegative functions f. For a general nonnegative function f, there exist a sequence $\{f_n\}_{n \geq 1}$ of nonnegative simple functions such that $f_n(\omega_1, \omega_2) \uparrow f(\omega_1, \omega_2)$ for all $(\omega_1, \omega_2) \in \Omega_1 \times \Omega_2$. Write $g_{1n}(\omega_1) = \int_{\Omega_1} f_n(\omega_1, \omega_2)\mu_2(d\omega_2)$. Then, g_{1n} is $\mathcal{F}_1$-measurable for all $n \geq 1$, g_{1n}'s are nondecreasing, and by the MCT,

$$\begin{aligned} g_1(\omega_1) &\equiv \int_{\Omega_1} f(\omega_1, \omega_2)\mu_2(d\omega_2) \\ &= \lim_{n\to\infty} \int f_n(\omega_1, \omega_2)\mu_2(d\omega_2) \\ &= \lim_{n\to\infty} g_{1n}(\omega_1) \qquad (2.6) \end{aligned}$$

for all $\omega_1 \in \Omega_1$. Thus, by Proposition 2.1.5, g_1 is $\langle \mathcal{F}_1, \mathcal{B}(\bar{\mathbb{R}}) \rangle$-measurable. Since (2.5) holds for simple functions, $\int f_n d\mu = \int g_{1n} d\mu_1$ for all $n \geq 1$. Hence, by repeated applications of the MCT, it follows that

$$\begin{aligned} \int f d\mu &= \lim_{n\to\infty} \int f_n d\mu = \lim_{n\to\infty} \int g_{1n} d\mu_1 \\ &= \int (\lim_{n\to\infty} g_{1n}) d\mu_1 = \int g_1 d\mu_1. \end{aligned}$$

The proofs of (2.4) and the second equality in (2.5) are similar. □

Theorem 5.2.2: (*Fubini's theorem*). *Let* $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$ *be* σ*-finite measure spaces and let* $f \in L^1(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, \mu_1 \times \mu_2)$. *Then there exist sets* $B_i \in \mathcal{F}_i$, $i = 1, 2$ *such that*

(i) $\mu_i(\Omega_i \setminus B_i) = 0$ *for* $i = 1, 2$,

(ii) for $\omega_1 \in B_1$, $f(\omega_1, \cdot) \in L^1(\Omega_2, \mathcal{F}_2, \mu_2)$,

(iii) the function

$$g_1(\omega_1) \equiv \begin{cases} \int_{\Omega_2} f(\omega_1, \omega_2)\mu_2(d\omega_2) & \text{for} \quad \omega_1 \text{ in } B_1 \\ 0 & \text{for} \quad \omega_1 \text{ in } B_1^c \end{cases}$$

is $\mathcal{F}_1$*-measurable and*

$$\int_{\Omega_1} g_1 d\mu_1 = \int_{\Omega_1 \times \Omega_2} f d(\mu_1 \times \mu_2), \qquad (2.7)$$

(iv) for $\omega_2 \in B_2$, $f(\cdot, \omega_2) \in L^1(\Omega_1, \mathcal{F}_1, \mu_1)$,

(v) the function

$$g_2(\omega_2) \equiv \begin{cases} \int_{\Omega_1} f(\omega_1, \omega_2)\mu_1(d\omega_1) & \textit{for} \quad \omega_2 \textit{ in } B_2 \\ 0 & \textit{for} \quad \omega_2 \textit{ in } B_2^c \end{cases}$$

is $\mathcal{F}_2$-measurable and

$$\int_{\Omega_2} g_2 d\mu_2 = \int_{\Omega_1 \times \Omega_2} f d(\mu_1 \times \mu_2). \tag{2.8}$$

Remark 5.2.1: An informal statement of the above theorem is as follows. If f is integrable on the product space, then the sectional integrals $\int_{\Omega_2} f(\omega_1, \cdot)d\mu_2$ and $\int_{\Omega_1} f(\cdot, \omega_2)d\mu_1$ are well defined a.e., and their integrals w.r.t. μ_1 and μ_2, respectively, are equal to the integral of f w.r.t. the product measure $\mu_1 \times \mu_2$.

Proof: By Tonelli's theorem

$$\int_{\Omega_1 \times \Omega_2} |f| d(\mu_1 \times \mu_2) = \int_{\Omega_1} \left(\int_{\Omega_2} |f(\omega_1, \omega_2)| \mu_2(d\omega_2) \right) \mu_1(d\omega_1).$$

So $\int_{\Omega_1 \times \Omega_2} |f| d(\mu_1 \times \mu_2) < \infty$ implies that $\mu_1(B_1^c) = 0$ where $B_1 = \{\omega_1 : \int |f(\omega_1, \cdot)| d\mu_2 < \infty\}$. Also, by Tonelli's theorem

$$g_{11}(\omega_1) \equiv \int_{\Omega_2} f^+(\omega_1, \cdot) d\mu_2 \quad \text{and} \quad g_{12}(\omega_1) \equiv \int_{\Omega_2} f^-(\omega_1, \cdot) d\mu_2$$

are both $\mathcal{F}_1$-measurable and

$$\int_{\Omega_1} g_{11} d\mu_1 = \int_{\Omega_1 \times \Omega_2} f^+ d(\mu_1 \times \mu_2), \quad \int_{\Omega_1} g_{12} d\mu_1 = \int_{\Omega_1 \times \Omega_2} f^- d(\mu_1 \times \mu_2). \tag{2.9}$$

Since g_1 defined in (iii) can be written as $g_1 = (g_{11} - g_{12}) I_{B_1}$, g_1 is $\mathcal{F}_1$-measurable. Also,

$$\begin{aligned} \int_{\Omega_1} |g_1| d\mu_1 &\leq \int_{\Omega_1} g_{11} d\mu_1 + \int_{\Omega_1} g_{12} d\mu_1 \\ &= \int_{\Omega_1 \times \Omega_2} f^+ d(\mu_1 \times \mu_2) + \int_{\Omega_1 \times \Omega_2} f^- d(\mu_1 \times \mu_2) \\ &< \infty. \end{aligned}$$

Further, as $\int_{\Omega_1 \times \Omega_2} |f| d\mu_1 \times \mu_2 < \infty$, by (2.9), g_{11} and $g_{12} \in L^1(\Omega_1, \mathcal{F}_1, \mu_1)$. Noting that $\mu_1(B_1^c) = 0$, one gets

$$\begin{aligned} \int_{\Omega_1} g_1 d\mu_1 &= \int_{\Omega_1} (g_{11} - g_{12}) I_{B_1} d\mu_1 \\ &= \int_{\Omega_1} g_{11} I_{B_1} d\mu_1 - \int_{\Omega_1} g_{12} I_{B_1} d\mu_1 \\ &= \int_{\Omega_1} g_{11} d\mu_1 - \int_{\Omega_1} g_{12} d\mu_1 \end{aligned}$$

which, by (2.9), equals $\int_{\Omega_1\times\Omega_2} f^+ d(\mu_1 \times \mu_2) - \int_{\Omega_1\times\Omega_2} f^- d(\mu_1 \times \mu_2) = \int_{\Omega_1\times\Omega_2} f d(\mu_1 \times \mu_2)$. Thus, (ii) and (iii) of the theorem have been established as well as (i) for $i = 1$. The proofs of (iv) and (v) and that of (i) for $i = 2$ are similar. $\square$

An application of the Fubini-Tonelli theorems gives an *integration by parts* formula. Let F_1 and F_2 be two nondecreasing right continuous functions on an interval $[a, b]$. Let μ_i be the Lebesgue-Stieltjes measure on $\mathcal{B}([a, b])$ corresponding to F_i, $i = 1, 2$. The 'integration by parts' formula allows one to write $\int_{(a,b]} F_1(x)dF_2(x) \equiv \int_{(a,b]} F_1 d\mu_2$ in terms of $\int_{(a,b]} F_2(x)dF_1(x) \equiv \int_{(a,b]} F_2 d\mu_1$.

Theorem 5.2.3: *Let F_1, F_2 be two nondecreasing right continuous functions on $[a, b]$ with no common points of discontinuity in $(a, b]$. Then*

$$\int_{(a,b]} F_1(x)dF_2(x) = F_1(b)F_2(b) - F_1(a)F_2(a) - \int_{(a,b]} F_2(x)dF_1(x). \tag{2.10}$$

Proof: Note that $\big((a, b], \mathcal{B}(a, b], \mu_i\big)$, $i = 1, 2$ are finite measure spaces. Consider the product space $\big((a, b] \times (a, b], \mathcal{B}((a, b] \times (a, b]), \mu_1 \times \mu_2\big)$. Define the sets

$$\begin{aligned} A &= \{(x, y) : a < x \le y \le b\} \\ B &= \{(x, y) : a < y \le x \le b\} \end{aligned}$$

and

$$C = \{(x, y) : a < x = y \le b\}.$$

For notational simplicity, write A_x and A_y for the x-section and the y-section of A, respectively, and similarly, for the sets B and C. Since F_1 and F_2 have no common points of discontinuity, by Theorem 5.1.2

$$\begin{aligned} \mu_1 \times \mu_2(C) &= \int_{(a,b]} \mu_2(C_x)\mu_1(dx) \\ &= \int_{(a,b]} \mu_2(\{x\})\mu_1(dx) \\ &= 0. \end{aligned} \tag{2.11}$$

(see Problem 5.3). And by Theorem 5.1.2,

$$\begin{aligned} \mu_1 \times \mu_2(A) &= \int_{(a,b]} \mu_1(A_y)\mu_2(dy) \\ &= \int_{(a,b]} \mu_1((a, y])\mu_2(dy) \\ &= \int_{(a,b]} [F_1(y) - F_1(a)]dF_2(y) \end{aligned} \tag{2.12}$$

and similarly,

$$\begin{aligned} \mu_1 \times \mu_2(B) &= \int_{(a,b]} \mu_2(B_x)\mu_1(dx) \\ &= \int_{(a,b]} [F_2(x) - F_2(a)]dF_1(x). \end{aligned} \tag{2.13}$$

Next note that $(\mu_1 \times \mu_2)((a,b] \times (a,b]) = \mu_1((a,b]) \cdot \mu_2((a,b]) = [F_1(b) - F_1(a)][F_2(b) - F_2(a)]$. Hence, by (2.11)–(2.13),

$$\begin{aligned} & [F_1(b) - F_1(a)][F_2(b) - F_2(a)] \\ = \;& (\mu_1 \times \mu_2)((a,b] \times (a,b]) \\ = \;& \mu_1 \times \mu_2(A) + \mu_1 \times \mu_2(B) - \mu_1 \times \mu_2(C) \\ = \;& \int_{(a,b]} [F_1(y) - F_1(a)]dF_2(y) + \int_{(a,b]} [F_2(x) - F_2(a)]dF_1(x), \end{aligned}$$

which yields (2.10), thereby completing the proof of Theorem 5.2.3. □

If F_1 and F_2 are absolutely continuous with nonnegative densities f_1 and f_2 w.r.t. the Lebesgue measure on $(a,b]$, then (2.10) yields

$$\int_a^b F_1(x)f_2(x)dx = F_1(b)F_2(b) - F_1(a)F_2(a) - \int_a^b F_2(x)f_1(x)dx. \tag{2.14}$$

If f_1 and f_2 are any two Lebesgue integrable function on $(a,b]$ that are not necessarily nonnegative, one can decompose f_i as $f_i = f_i^+ - f_i^-$ and apply (2.14) to f_i^+'s and f_i^-'s separately. Then, by linearity, it follows that the relation (2.14) also holds for the given f_1 and f_2. Thus, the standard 'integration by parts' formula is a special case of Theorem 5.2.3. Relations (2.10) and (2.14) can be extended to unbounded intervals under suitable conditions on F_1 and F_2 (Problem 5.5).

Remark 5.2.2: The measure space $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2, \mu_1 \times \mu_2)$ constructed using the integrals in (1.4) and (1.5) is not necessarily complete. As mentioned at the beginning of this section, the approach based on the extension theorem (applied to the set function defined in (1.1) on the algebra $\mathcal{A}$ of finite disjoint unions of measurable rectangles) does yield a complete measure space $(\Omega_1 \times \Omega_2, \mathcal{M}, \lambda)$ such that $\mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{M}$ and $\lambda(A) = (\mu_1 \times \mu_2)(A)$ for all A in $\mathcal{F}_1 \times \mathcal{F}_2$. To compute the integral $\int f d\lambda$ of an $\mathcal{M}$-measurable function f w.r.t. λ, formula (2.5) in Tonelli's theorem and formulas (2.7) and (2.8) in Fubini's theorem continue to be valid with some modifications. In the following, a modified statement of Fubini's theorem is presented. Similar modification also holds for Tonelli's theorem (cf. Royden (1988), Chapter 12, Section 4).

Theorem 5.2.4: *Let $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$ be σ-finite measure spaces and let $f \in L^1(\Omega_1 \times \Omega_2, \mathcal{M}, \mu_1 \times \mu_2)$. Let $(\Omega_i, \bar{\mathcal{F}}_i, \bar{\mu}_i)$ be the completion of $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, 2$. Then there exist sets B_i in $\bar{\mathcal{F}}_i$, $i = 1, 2$ such that*

(i) $\bar{\mu}_i(B_i^c) = 0$, $i = 1, 2$,

(ii) for $\omega_1 \in B_1$, $f(\omega_1, \cdot)$ *is* $\bar{\mathcal{F}}_2$*-measurable,*

(iii) $\int_{\Omega_2} |f(\omega_1, \cdot)| d\bar{\mu}_2 < \infty$ *for all* $\omega_1 \in B_1$,

(iv) $g_1(\omega_1) \equiv \begin{cases} \int_{\Omega_2} f(\omega_1, \cdot) d\bar{\mu}_2 & \textit{for} \quad \omega_1 \textit{ in } B, \\ 0 & \textit{for} \quad \omega_1 \textit{ in } B_1^c \end{cases}$
is $\bar{\mathcal{F}}_1$*-measurable,*

(v) $\int_{\Omega_1} g_1 d\bar{\mu}_1 = \int_{\Omega_1 \times \Omega_2} f d\lambda$.

Further, a similar statement holds for $i = 2$.

5.3 Extensions to products of higher orders

Definition 5.3.1: Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, \ldots, k$ be measurable spaces, $2 \leq k < \infty$. Then

(a) $\times_{i=1}^k \Omega_i = \Omega_1 \times \ldots \times \Omega_k = \{(\omega_1, \ldots, \omega_i) : \omega :\in \Omega_i$ for $i = 1, \ldots, k\}$ is called the *product set* of $\Omega_1, \ldots, \Omega_k$.

(b) A set of the form $A_1 \times \ldots \times A_k$ with $A_i \in \mathcal{F}_i, 1 \leq i \leq k$ is called a *measurable rectangle*. The *product σ-algebra* on $\times_{i=1}^k \Omega_i$, denoted by $\times_{i=1}^k \mathcal{F}_i$, is the σ-algebra generated by the collection of all measurable rectangles, i.e.,

$$\times_{i=1}^k \mathcal{F}_i = \sigma\langle\{A_1 \times \ldots \times A_k : A_i \in \mathcal{F}_i, 1 \leq i \leq k\}\rangle.$$

(c) $(\times_{i=1}^k \Omega_i, \times_{i=1}^k \mathcal{F}_i)$ is called the *product space* or the *product measurable space*. When $(\Omega_i, \mathcal{F}_i) = (\Omega, \mathcal{F})$ for all $1 \leq i \leq k$, the product space will be denoted by $(\Omega^k, \mathcal{F}^k)$.

To define the product measure, one starts with a natural set function on $\mathcal{C}$, the class of all measurable rectangles, extends it to the algebra $\mathcal{A}$ of finite disjoint unions of measurable rectangles by linearity, and then uses the extension theorem (Theorem 1.3.3) to obtain a measure on the product space. The details of this construction are now described below.

Define a set function μ on $\mathcal{C}$ by

$$\mu(A) = \prod_{i=1}^k \mu(A_i) \tag{3.1}$$

where $A = A_1 \times A_2 \times \quad \times A_k$ with $A_i \in \mathcal{F}_i$, $i = 1, 2, \ldots, k$. Next extend it to the algebra $\mathcal{A}$ by linearity. If $B \in \mathcal{A}$ is of the form $B = \bigcup_{j=1}^m B_j$ where

$\{B_j : 1, \ldots, m\} \subset \mathcal{C}$ are disjoint, then define $\mu(B)$ by

$$\mu(B) = \sum_{j=1}^{m} \mu(B_j). \tag{3.2}$$

If the set $B \in \mathcal{A}$ admits two different representations as finite unions of measurable rectangles then it is not difficult to verify that the value of $\mu(B)$ remains unchanged. Next it is shown that μ is a measure on $\mathcal{A}$.

Proposition 5.3.1:. *Let μ be as defined by (3.1) and (3.2) above on the algebra $\mathcal{A}$. Then, it is countably additive on $\mathcal{A}$.*

Proof: Let $\{B_n\}_{n=1}^{\infty} \subset \mathcal{A}$ be disjoint such that $B = \bigcup_{n\geq 1} B_n$ also belongs to $\mathcal{A}$. It is enough to show that

$$\mu(B) = \sum_{n=1}^{\infty} \mu(B_n). \tag{3.3}$$

Let B and $\{B_n\}_{n=1}^{\infty}$ admit the representations $B = \bigcup_{i=1}^{\ell} A_i$ and $B_j = \bigcup_{r=1}^{\ell_j} A_{jr}$ where ℓ and ℓ_j are (finite) integers, A_i, $A_{jr} \in \mathcal{C}$ and each of the collections $\{A_i\}_{i=1}^{\ell}$ and $\{A_{jr}\}_{r=1}^{\ell_j}$, $j \geq 1$ is disjoint. Then each A_i can be written as

$$A_i = A_i \cap \Big[\bigcup_{n\geq 1} B_n\Big] = \bigcup_{n\geq 1} \bigcup_{r=1}^{\ell_n} (A_i \cap A_{nr}).$$

Suppose it is shown that for each $i \geq 1$,

$$\mu(A_i) = \sum_{j=1}^{\infty} \sum_{r=1}^{\ell_j} \mu(A_i \cap A_{jr}). \tag{3.4}$$

Then, by (3.2),

$$\begin{aligned}
\mu(B) &= \sum_{i=1}^{\ell} \mu(A_i) \\
&= \sum_{i=1}^{\ell} \sum_{j=1}^{\infty} \sum_{r=1}^{\ell_j} \mu(A_i \cap A_{jr}) \\
&= \sum_{j=1}^{\infty} \sum_{r=1}^{\ell_j} \sum_{i=1}^{\ell} \mu(A_i \cap A_{jr}) \\
&= \sum_{j=1}^{\infty} \sum_{r=1}^{\ell_j} \mu(B \cap A_{jr}),
\end{aligned}$$

where the last equality follows from the representation of $B \cap A_{jr}$ as $\bigcup_{i=1}^{\ell}(A_i \cap A_{jr})$ and the fact that $A_i \cap A_{jr} \in \mathcal{C}$ for all i, j, r. Since $A_{jr} \subset B_j \subset B$, the above yields

$$\sum_{j=1}^{\infty} \sum_{r=1}^{\ell_j} \mu(A_{jr}) = \sum_{j=1}^{\infty} \mu(B_j) \qquad \text{(by (3.2))}$$

which establishes (3.3).

Thus, it remains to show (3.4). This is implied by the following:
Let $C = \bigcup_{n\geq 1} C_n$ where $\{C_n\}_{n=1}^{\infty}$ is a collection of disjoint measurable rectangles and C is also a measurable rectangle. Then

$$\mu(C) = \sum_{i=1}^{\infty} \mu(C_i). \tag{3.5}$$

Let $C = A_1 \times A_2 \times \quad \times A_k$ and $C_i = A_{i1} \times A_{i2} \times \quad \times A_{ik}$, $i = 1, 2, \ldots$. Since $C = \bigcup_{n\geq 1} C_n$ and $\{C_n\}_{n=1}^{\infty}$ are disjoint, this implies $I_C(\omega_1, \ldots, \omega_k) = \sum_{i=1}^{\infty} I_{C_i}(\omega_1, \ldots, \omega_k)$, for all $(\omega_1, \ldots, \omega_k) \in \Omega_1 \times \ldots \times \Omega_k$. That is,

$$\prod_{j=1}^{k} I_{A_j}(\omega_j) = \sum_{i=1}^{\infty} \prod_{j=1}^{k} I_{A_{ij}}(\omega_j) \tag{3.6}$$

for all $(\omega_1, \ldots, \omega_k) \in \Omega_1 \times \ldots \times \Omega_k$. Integration of both sides of (3.6) w.r.t. μ_1 over Ω_1 yields

$$\mu_1(A_1)\Big(\prod_{j=2}^{k} I_{A_j}(\omega_j)\Big) = \sum_{i=1}^{\infty} \mu_1(A_{i1}) \prod_{j=2}^{k} I_{A_{ij}}(\omega_j),$$

and by iteration

$$\prod_{j=1}^{k} \mu_j(A_j) = \sum_{i=1}^{\infty} \prod_{j=1}^{k} \mu_j(A_{ij}).$$

Hence, (3.5) follows. $\square$

By the extension theorem (Theorem 1.3.3), there exists a σ-algebra $\mathcal{M}$ and a measure $\bar{\mu}$ such that

(i) $(\times_{i=1}^{k}\Omega_i, \mathcal{M}, \bar{\mu})$ is complete,

(ii) $\times_{i=1}^{k}\mathcal{F}_i \subset \mathcal{M}$, and

(iii) $\bar{\mu}(A) = \mu(A)$ for all $A \in \mathcal{A}$.

Thus, the above procedure yields an extension $\bar{\mu}$ of μ on $\mathcal{A}$, and this extension is unique when all the μ_i's are σ-finite. Further, under the hypothesis that $\mu_1, \ldots, \mu_k$ are σ-finite, the analogs of formulas (1.4) and (1.5) for computing the product measure of a set via the iterated integrals are valid. More generally, the Tonelli-Fubini theorems extend to the k-fold product spaces in an obvious manner. For example, if $(\Omega_i, \mathcal{F}_i, \mu_i)$ are σ-finite measure spaces for $i = 1, \ldots, k$ and f is a nonnegative measurable function on $(\times_{i=1}^k \Omega_i, \times_{i=1}^k \mathcal{F}_i, \bar{\mu})$, then

$$\int f d\bar{\mu} = \int_{\Omega_{i_1}} \int_{\Omega_{i_2}} \cdots \int_{\Omega_{i_k}} f(\omega_1, \omega_2, \quad \omega_k) \mu_{i_1}(d\omega_{i_1}) \ldots \mu_{i_k}(d\omega_{i_k})$$

for any permutation $(i_1, i_2, \ldots, i_k)$ of $(1, 2, \ldots, k)$.

Definition 5.3.2: The measure space $(\times_{i=1}^k \Omega_i, \mathcal{M}, \bar{\mu})$ is called the *complete product measure space* and $\bar{\mu}$ the *complete product measure.*

Remark 5.3.1: If $(\Omega_i, \mathcal{F}_i, \mu_i) \equiv (\mathbb{R}, \mathcal{L}, m)$ where $\mathcal{L}$ is the Lebesgue σ-algebra and m is the Lebesgue measure, then the above extension coincides with the $(\mathbb{R}^k, \mathcal{L}^k, m^k)$, where $\mathcal{L}^k$ is the Lebesgue σ-algebra in $\mathbb{R}^k$ and m^k is the k dimensional Lebesgue measure.

5.4 Convolutions

5.4.1 Convolution of measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

In this section, convolution of measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ is discussed. From this, one can easily obtain convolution of sequences, convolution of functions in $L^1(\mathbb{R})$ and convolution of functions with measures.

Proposition 5.4.1: *Let μ and λ be two σ-finite measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. For any Borel set A in $\mathcal{B}(\mathbb{R})$, let*

$$(\mu * \lambda)(A) \equiv \int \int I_A(x + y) \mu(dx) \lambda(dy). \tag{4.1}$$

Then $(\mu * \lambda)(\cdot)$ is a measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$.

Proof: Let $h(x, y) \equiv x + y$ for $(x, y) \in \mathbb{R} \times \mathbb{R}$. Then $h : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is continuous and hence is $\langle \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$-measurable. Consider the measure $(\mu \times \lambda) h^{-1}(\cdot)$ on $\langle \mathbb{R}, \mathcal{B}(\mathbb{R}) \rangle$ induced by the map h and the measure $\mu \times \lambda$ on $\langle \mathbb{R} \times \mathbb{R}, \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) \rangle$. Clearly

$$(\mu * \lambda)(\cdot) = (\mu \times \lambda) h^{-1}(\cdot)$$

and hence the proposition is proved. $\square$

Definition 5.4.1: For any two σ-finite measures μ and λ on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$, the measure $(\mu * \lambda)(A)$ defined in (4.1) is called the *convolution of μ and λ.*

The following proposition is easy to verify.

Proposition 5.4.2: *Let μ_1, μ_2, μ_3 be σ-finite measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Then*

(i) (*Commutativity*) $\mu_1 * \mu_2 = \mu_2 * \mu_1$,

(ii) (*Associativity*) $\mu_1 * (\mu_2 * \mu_3) = (\mu_1 * \mu_2) * \mu_3$,

(iii) (*Distributive*) $\mu_1 * (\mu_2 + \mu_3) = \mu_1 * \mu_2 + \mu_1 * \mu_3$,

(iv) (*Identity element*) $\mu_1 * \delta_{\{0\}} = \mu_1$

where $\delta_{\{0\}}(\cdot)$ is the delta measure at 0, i.e.,

$$\delta_{\{0\}}(A) = \begin{cases} 1 & \textit{if} \quad 0 \in A, \\ 0 & \textit{if} \quad 0 \notin A \end{cases}$$

for all $A \in \mathcal{B}(\mathbb{R})$.

Remark 5.4.1: (*Extension to* $\mathbb{R}^k$). Definition 5.4.1 extends to measures on $\big(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\big)$ for any integer $k \geq 1$ as well as to any space that is a commutative group under addition such as the sequence space $\mathbb{R}^\infty$, the function spaces $\mathcal{C}[0,1]$ and $\mathcal{C}[0,\infty)$. These are relevant in the study of stochastic processes.

Remark 5.4.2: (*Sums of independent random variables*). It will be seen in Chapter 7 that if X and Y are two independent random variables on a probability space $(\Omega, \mathcal{F}, P)$, then $P_X * P_Y = P_{X+Y}$ where for any random variable Z on $(\Omega, \mathcal{F}, P)$, P_Z is the distribution of Z, i.e., the probability measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ induced by P and Z, i.e., $P_Z(\cdot) = PZ^{-1}(\cdot)$ on $\mathcal{B}(\mathbb{R})$.

Remark 5.4.3: (*Extension to signed measures*). Let μ and λ be two finite signed measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ as defined in Section 4.2. Let $\mu = \mu^+ - \mu^-$, $\lambda = \lambda^+ - \lambda^-$ be the Jordan decomposition of μ and λ, respectively. Then the *product measure* $\mu \times \lambda$ can be defined as the signed measure

$$\begin{aligned} \mu \times \lambda &\equiv \big[(\mu^+ \times \lambda^+) + (\mu^- \times \lambda^-)\big] - \big[(\mu^+ \times \lambda^-) + (\mu^- + \lambda^+)\big] \\ &\equiv \gamma_1 - \gamma_2, \quad \text{say.} \end{aligned} \tag{4.2}$$

This is well defined since the measures $(\mu^+ \times \lambda^+) + (\mu^- \times \lambda^-)$ and $(\mu^+ \times \lambda^-) + (\mu^- \times \lambda^+)$ are both finite measures. Now the definition of convolution of measures in (4.1) carries over to signed measures using the definition of integration w.r.t. signed measures discussed in Section 4.2.

Definition 5.4.2: Let μ and ν be (finite) signed measures on a measurable space $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. The *convolution of μ and ν*, denoted by $\mu * \nu$ is the signed measure defined by

$$(\mu * \nu)(A) \quad = \quad \int\int I_A(x+y) d(\mu \times \nu)$$

$$= \int\int I_A(x+y)d\gamma_1 - \int\int I_A(x+y)d\gamma_2 \qquad (4.3)$$

where γ_1 and γ_2 are as in (4.2).

5.4.2 Convolution of sequences

Definition 5.4.3: Let $\tilde{a} \equiv \{a_n\}_{n\geq 0}$ and $\tilde{b} \equiv \{b_n\}_{n\geq 0}$ be two sequences of real numbers. Then the *convolution of* $\tilde{a}$ *and* $\tilde{b}$ denoted by $\tilde{a} * \tilde{b}$ is the sequence

$$(\tilde{a} * \tilde{b})(n) = \sum_{j=0}^{n} a_j b_{n-j}, \quad n \geq 0. \qquad (4.4)$$

If $\sum_{j=0}^\infty |a_j| < \infty$ and $\sum_{j=0}^\infty |b_j| < \infty$, then $\tilde{a} * \tilde{b}$ corresponds to the convolution of the signed measures $\tilde{a}$ and $\tilde{b}$ on $\mathbb{Z}_+$, defined by $\tilde{a}(i) = a_i$, $\tilde{b}(i) = b_i$, $i \in \mathbb{Z}_+$.

Example 5.4.1:

(a) Let $b_n \equiv 1$ for $n \geq 0$. Then for any $\{a_n\}_{n\geq 0}$, $(\tilde{a} * \tilde{b})(n) = \sum_{j=0}^n a_j$.

(b) Fix $0 < p < 1$, k_1, k_2 positive integers. For $n \in \mathbb{Z}_+$, let

$$\begin{aligned} a_n &\equiv \binom{k_1}{n} p^n (1-p)^{k_1-n} I_{[0,k_1]}(n) \\ b_n &\equiv \binom{k_2}{n} p^n (1-p)^{k_2-n} I_{[0,k_2]}(n). \end{aligned}$$

Then $(\tilde{a} * \tilde{b})(n) = \binom{k_1+k_2}{n} p^n (1-p)^{k_2-n} I_{[0,k_1+k_2]}(n)$.

(c) Fix $0 < \lambda_1, \lambda_2 < \infty$. Let

$$\begin{aligned} a_n &\equiv e^{-\lambda_1} \frac{\lambda_1^n}{n!}, \ n = 0, 1, 2, \ldots \\ b_n &\equiv e^{-\lambda_2} \frac{\lambda_2^n}{n!}, \ n = 0, 1, 2, \ldots \ . \end{aligned}$$

Then $(\tilde{a} * \tilde{b})(n) = e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1+\lambda_2)^n}{n!}$, $n = 0, 1, 2, \ldots$.

The verification of these claims is left as an exercise (Problem 5.20).

A useful technique to determine $\tilde{a} * \tilde{b}$ is the use of generating functions. This will be discussed in Section 5.5.

5.4.3 Convolution of functions in $L^1(\mathbb{R})$

Let $L^1(\mathbb{R}) \equiv L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ where $m(\cdot)$ is the Lebesgue measure. In the following, for $f \in L^1(\mathbb{R})$, $\int f dm$ will also be written as $\int f(x)dx$.

Proposition 5.4.3: *Let f, $g \in L^1(\mathbb{R})$. Then for almost all x (w.r.t. m),*

$$\int |f(x-u)|\,|g(u)|du < \infty. \tag{4.5}$$

Proof: Let $k(x,u) = f(x-u)g(u)$. Since $\pi(x,u) \equiv x-u$ is a continuous map from $\mathbb{R}\times\mathbb{R} \to \mathbb{R}$, it is $\langle\mathcal{B}(\mathbb{R})\times\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable. Also, since f and g are Borel measurable, it follows that $k : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ is $\langle\mathcal{B}(\mathbb{R})\times\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable. Also note that by the translation invariance of m, $\int |f(x-u)|dx = \int |f(x)|dx$ for any Borel measurable $f : \mathbb{R} \to \mathbb{R}$. Hence, by Tonelli's theorem,

$$\begin{aligned}\int\int |k(x,u)|dxdu &= \int\Big(\int |k(x,u)|dx\Big)du \\ &= \int |g(u)|\Big(\int |f(x)|dx\Big)du \\ &= \|g\|_1\,\|f\|_1 < \infty. \end{aligned} \tag{4.6}$$

Also by Tonelli's theorem

$$\int\int |k(x,u)|dxdu = \int\Big(\int |k(x,u)|du\Big)dx.$$

By (4.6), this yields

$$\int |k(x,u)|du < \infty \quad \text{a.e.} \quad (m)$$

which is the same as (4.5). □

Definition 5.4.4: Let f, $g \in L^1(\mathbb{R})$. Then the *convolution of f and g*, denoted by $f*g$ is the function defined a.e. (m) by

$$(f*g)(x) \equiv \int f(x-u)g(u)du. \tag{4.7}$$

Note that by (4.5) this is well defined.

Proposition 5.4.4: *Let f, $g \in L^1(\mathbb{R})$. Then*

(i) $f*g = g*f$

(ii) $f*g \in L^1(\mathbb{R})$ *and* $\|f*g\|_1 \le \|f\|_1\,\|g\|_1$

(iii) $\int f*g\,dm = \Big(\int f dm\Big)\Big(\int g dm\Big)$.

Proof: For (i), use the change of variable $u \to x-u$ and the translation invariance of m. For (ii) use (4.6). For (iii), use Fubini's theorem. □

It is easy to see that if μ and λ denote the signed measures with Radon-Nikodym derivatives f and g, respectively, w.r.t. m, then $\mu * \lambda$ is the signed measure with Radon-Nikodym derivative $f * g$ w.r.t. m.

Example 5.4.2: Here are two examples from probability theory.

(a) (*Convolutions of Uniform* $[0,1]$ *densities*). Let $f = g = I_{[0,1]}$. Then

$$(f * g)(x) = \begin{cases} x & 0 \le x \le 1 \\ 2 - x & 1 \le x \le 2. \end{cases}$$

(b) (*Convolutions of* $N(0,1)$ *densities*). Let $f(x) \equiv g(x) \equiv \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $-\infty < x < \infty$. Then $(f * g)(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2}} e^{-\frac{x^2}{4}}$, $-\infty < x < \infty$.

5.4.4 Convolution of functions and measures

Let $f \in L^1(\mathbb{R})$ and λ be a signed measure. Let $\mu(A) \equiv \int_A f dm$, $A \in \mathcal{B}(\mathbb{R})$. Then it can be shown that $\mu * \lambda$ is dominated by m and its Radon-Nikodym derivative is given by

$$f * \lambda(x) \equiv \int f(x-u)\lambda(du), \quad x \in \mathbb{R}. \tag{4.8}$$

Note that $f * \lambda$ in (4.8) is well defined for any nonnegative Borel measurable f and any measure λ on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$.

5.5 Generating functions and Laplace transforms

Definition 5.5.1: Let $\{a_n\}_{n\ge0}$ be a sequence in $\mathbb{R}$ and let $\rho = \big(\overline{\lim_{n\to\infty}} |a_n|^{1/n}\big)^{-1}$. The power series

$$A(s) \equiv \sum_{n=0}^{\infty} a_n s^n, \tag{5.1}$$

defined for all s in $(-\rho, \rho)$, is called the *generating function of* $\{a_n\}_{n\ge0}$.

Note that the power series in (5.1) converges absolutely for $|s| < \rho$ and ρ is called the *radius of convergence* of $A(s)$ (cf. Appendix A.2).

The usefulness of generating functions is given by the following:

Proposition 5.5.1: *Let* $\{a_n\}_{n\ge0}$ *and* $\{b_n\}_{n\ge0}$ *be two sequences in* $\mathbb{R}$. *Let* $\{c_n\}_{n\ge0}$ *be the convolution of* $\{a_n\}$ *and* $\{b_n\}$. *That is,*

$$c_n = \sum_{j=0}^{n} a_j b_{n-j}, \ n \ge 0.$$

Then

$$C(s) = A(s)B(s) \tag{5.2}$$

for all s in $(-\rho, \rho)$, where $A(\cdot)$, $B(\cdot)$, and $C(\cdot)$ are, respectively, the generating function of the sequences $\{a_n\}_{n\geq 0}$, $\{b_n\}_{n\geq 0}$ and $\{c_n\}_{n\geq 0}$ and $\rho = \min(\rho_a, \rho_b)$ with $\rho_a = \big(\overline{\lim}_{n\to\infty} |a_n|^{1/n}\big)^{-1}$ and $\rho_b = \big(\overline{\lim}_{n\to\infty} |b_n|^{1/n}\big)^{-1}$.

Proof: By Tonelli's theorem applied to the counting measure on the product space $(\mathbb{Z}_+ \times \mathbb{Z}_+)$,

$$\begin{aligned} & \sum_{n=0}^{\infty} |s|^n \Big(\sum_{j=0}^{n} |a_j|\,|b_{n-j}| \Big) \\ = \quad & \sum_{j=0}^{\infty} |a_j|\,|s|^j \Big(\sum_{n=j}^{\infty} |s|^{n-j} |b_{n-j}| \Big) \\ = \quad & A(|s|)B(|s|) < \infty \quad \text{if} \quad |s| < \rho. \end{aligned}$$

Now by Fubini's theorem, (5.2) follows. □

It is easy to verify the claims in Example 5.4.1 by using the above proposition and the uniqueness of the power series coefficients (Problem 5.20).

Proposition 5.5.2: (*Renewal sequences*). *Let $\{a_n\}_{n\geq 0}$, $\{b_n\}_{n\geq 0}$, $\{p_n\}_{n\geq 0}$ be sequences in $\mathbb{R}$ such that*

$$a_n = b_n + \sum_{j=0}^{n} a_{n-j} p_j, \ n \geq 0 \tag{5.3}$$

and $p_0 = 0$. Then

$$A(s) = \frac{B(s)}{1 - P(s)}$$

for all s such that $|s| < \rho$ and $P(s) \neq 1$, where $\rho = \min(\rho_b, \rho_p)$, $\rho_b = \big(\overline{\lim}_{n\to\infty} |b_n|^{1/n}\big)^{-1}$, $\rho_p = \big(\overline{\lim}_{n\to\infty} |p_n|^{1/n}\big)^{-1}$.

For applications of this to renewal theory, see Chapter 8.

Definition 5.5.2: (*Laplace transform*). Let $f : [0, \infty) \to \mathbb{R}$ be Borel measurable. The function

$$(Lf)(s) \equiv \int_{[0,\infty)} e^{sx} f(x) dx, \tag{5.4}$$

defined for all s in $\mathbb{R}$ such that

$$\int_{[0,\infty)} e^{sx} |f(x)| dx < \infty, \tag{5.5}$$

is called the *Laplace transform of f*.

It is easily seen that if (5.5) holds for some $s = s_0$, then it does for all $s < s_0$. The analog of Proposition 5.5.1 is the following.

Proposition 5.5.3: *Let f, $g \in L^1(\mathbb{R})$. Then*

$$L(f * g)(s) = Lf(s)Lg(s) \tag{5.6}$$

for all s such that (5.5) holds for both f and g.

Definition 5.5.3: (*Laplace-Stieltjes transform*). Let μ be a measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ such that $\mu\big((-\infty, 0)\big) = 0$. The function

$$\mu^*(s) \equiv L\mu(s) \equiv \int_{[0,\infty)} e^{sx}\mu(dx)$$

is called the *Laplace-Stieltjes transform* of μ. Clearly, $\mu^*(s)$ is well defined for all s in $\mathbb{R}$. However, $\mu^*(s_0) = \infty$ implies $\mu^*(s) = \infty$ for $s \geq s_0$.

Proposition 5.5.4: *Let μ and λ be measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$ such that $\mu\big((-\infty, 0)\big) = 0 = \lambda\big((-\infty, 0)\big)$. Then*

$$L(\mu * \lambda)(s) = L\mu(s)L\lambda(s) \ \ \textit{for all} \ \ s \text{ in } \mathbb{R}.$$

For an inversion formula to obtain a probability measure μ from $L\mu(\cdot)$, see Feller (1966), Section 13.4.

5.6 Fourier series

In this section, $L^p[0, 2\pi]$ stands for $L^p\big([0, 2\pi], \mathcal{B}([0, 2\pi]), m\big)$ where $m(\cdot)$ is Lebesgue measure and $0 < p < \infty$.

Definition 5.6.1: For $f \in L^1[0, 2\pi]$, $n \geq 0$, let

$$\begin{aligned} a_n &\equiv \frac{1}{2\pi}\int_0^{2\pi} f(x)\cos nx\, dx \\ b_n &\equiv \frac{1}{2\pi}\int_0^{2\pi} f(x)\sin nx\, dx, \end{aligned} \tag{6.1}$$

$$s_n(f, x) \equiv a_0 + \sum_{j=1}^{n}(a_j \cos jx + b_j \sin jx). \tag{6.2}$$

Then $\{a_n, b_n, n = 0, 1, 2, \ldots\}$ are called the *Fourier coefficients of f* and the sequence $\{s_n(f, x) : n = 0, 1, 2, \ldots\}$ is called the *partial sum sequence* of the *Fourier series expansion of f*.

Since $\mathcal{C}[0, 2\pi] \subset L^2[0, 2\pi] \subset L^1[0, 2\pi]$, the Fourier coefficients and the partial sum sequence of the Fourier series are well defined for f in $\mathcal{C}[0, 2\pi]$

and f in $L^2[0, 2\pi]$. J. Fourier introduced these series in the early 19th century to approximate certain functions exhibiting a periodic behavior that arose in the study of physics and mechanics and in particular in the theory of heat conduction (see Körner (1989) and Bhatia (2003)).

A natural question is: *In what sense does $s_n(f, x)$ approximate $f(x)$?* It turns out that one can prove the strongest results for f in $\mathcal{C}[0, 2\pi]$, less stronger results for f in $L^2[0, 2\pi]$, and finally, for f in $L^1[0, 2\pi]$.

In the early 19th century it was believed that for any f in $\mathcal{C}[0, 2\pi]$, $s_n(f, x)$ converged to $f(x)$ for all x in $[0, 2\pi]$. But in 1876, D. Raymond constructed a continuous function f for which $\overline{\lim}_{n\to\infty}|s_n(f; x_0)| = \infty$ for some x_0. However, Fejer showed in 1903 that for f in $\mathcal{C}[0, 2\pi]$, $s_n(f, \cdot)$ does converge to f uniformly in the *Cesaro sense*.

Theorem 5.6.1: (*Fejer*). *Let $f \in \mathcal{C}[0, 2\pi]$ and $s_n(f, \cdot)$, $n \geq 0$ be as in Definition 3.4.1. Let*

$$\mathcal{D}_n(f, \cdot) = \frac{1}{n} \sum_{j=0}^{n-1} s_j(f, \cdot), \ n \geq 1. \tag{6.3}$$

Then

$$\mathcal{D}_n(f, \cdot) \to f(\cdot) \quad \textit{uniformly on} \quad [0, 2\pi] \quad \textit{as} \quad n \to \infty. \tag{6.4}$$

For the proof of this result, the following result of some independent interest is needed.

Lemma 5.6.2: (*Fejer*). *Let for $m \geq 1$*

$$K_m(x) \equiv 1 + \frac{2}{m} \sum_{j=1}^{m-1} (m - j) \cos jx, \ x \in \mathbb{R}. \tag{6.5}$$

Then

(i)

$$K_m(x) = \begin{cases} \frac{1}{m}\left(\frac{\sin \frac{mx}{2}}{\sin \frac{x}{2}}\right)^2 & \text{if} \quad x \neq 0 \\ m & \text{if} \quad x = 0\ . \end{cases} \tag{6.6}$$

(ii) For $\delta > 0$,

$$\sup\{K_m(x) : \delta \leq |x| \leq 2\pi - \delta\} \to 0 \quad \textit{as} \quad m \to \infty. \tag{6.7}$$

(iii)

$$\frac{1}{2\pi} \int_0^{2\pi} K_m(x) dx = 1. \tag{6.8}$$

Proof: Clearly, (iii) follows from (6.5) on integration. Also, (ii) follows from (i) since for $\delta \leq x \leq 2\pi - \delta$,

$$K_m(x) \leq \frac{1}{m}\frac{1}{(\sin\frac{\delta}{2})^2} \to 0 \quad \text{as} \quad m \to \infty.$$

To establish (i) note that using Euler's formula (cf. Section A.3, Appendix) $e^{\iota x} = \cos x + \iota \sin x$, $K_m(x)$ can be written as

$$\begin{aligned} K_m(x) &= \frac{1}{m}\sum_{j=-(m-1)}^{(m-1)} (m-|j|)e^{\iota j x} \\ &= \frac{1}{m}\sum_{j=0}^{2(m-1)} (m-|j-(m-1)|)e^{\iota(j-(m-1))x} \\ &= \frac{1}{m}\Big(\sum_{k=0}^{m-1} e^{\iota(k-\frac{m-1}{2})x}\Big)^2. \end{aligned}$$

For $x \neq 0$,

$$\begin{aligned} K_m(x) &= \frac{1}{m}\Big(e^{-\frac{\iota(m-1)x}{2}}\frac{1-e^{\iota m x}}{1-e^{\iota x}}\Big)^2, \\ &= \frac{1}{m}\Big(\frac{e^{-\frac{\iota m x}{2}} - e^{\frac{\iota m x}{2}}}{e^{-\frac{\iota x}{2}} - e^{\frac{\iota x}{2}}}\Big)^2, \\ &= \frac{1}{m}\Big(\frac{\sin\frac{mx}{2}}{\sin\frac{x}{2}}\Big)^2. \end{aligned}$$

For $x = 0$, (6.5) yields

$$\begin{aligned} K_m(0) &= 1 + \frac{2}{m}\sum_{j=1}^{m-1}(m-j) \\ &= 1 + \frac{2}{m}\frac{(m-1)m}{2} = m. \end{aligned}$$

□

Proof of Theorem 5.6.1: Let $f \in \mathcal{C}[0, 2\pi]$. Let $\{a_n, b_n\}_{n\geq 0}$, $s_n(f,\cdot)$, $\mathcal{D}_m(f,\cdot)$ be as in (6.1), (6.2), and (6.3). Then from the definition of $a_n, b_n, n \geq 0$, it follows that

$$\begin{aligned} \mathcal{D}_m(f,\cdot) &= \frac{1}{2\pi}\int_0^{2\pi} K_m(x-u)f(u)du \\ &= \frac{1}{2\pi}\int_0^{2\pi} f(x-u)K_m(u)du \qquad (6.9) \end{aligned}$$

where $K_m(\cdot)$ is defined in (6.5) and $f(\cdot)$ is extended to all of $\mathbb{R}$ periodically with period 2π. Since $f \in \mathcal{C}[0, 2\pi]$, given $\epsilon > 0$, there exists a $\delta > 0$ such that

$$x, y \in [0, 2\pi],\ |x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon.$$

Also from (6.7), for this $\delta > 0$, there exist an m_0 such that $m \geq m_0 \Rightarrow \sup\{K_m(x) : \delta \leq x \leq 2\pi - \delta\} < \epsilon$. Now (6.9) yields, for $x \in [0, 2\pi]$, $m \geq m_0$

$$\begin{aligned} |\mathcal{D}_m(f, x) - f(x)| &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(x - u) - f(x)| K_m(u) du \\ &= \frac{1}{2\pi} \int_{(\delta \leq u \leq 2\pi - \delta)^c} |f(x - u) - f(x)| K_m(u) du \\ &\quad + \frac{1}{2\pi} \int_{(\delta \leq u \leq 2\pi - \delta)} |f(x - u) - f(x)| K_m(u) du \\ &\leq \frac{1}{2\pi} (\epsilon + 2\|f\| \epsilon 2\pi) \end{aligned}$$

where $\|f\| = \sup\{|f(x)| : x \in [0, 2\pi]\}$. Thus, $m \geq m_0 \Rightarrow \sup\{|\mathcal{D}_m(f, x) - f(x)| : x \in [0, 2\pi]\} \leq \epsilon \frac{(1 + 4\pi\|f\|)}{2\pi}$. Since $\epsilon > 0$ is arbitrary, the proof of Theorem 5.6.1 is complete. $\square$

An immediate consequence of Theorem 5.6.1 is the completeness of the trigonometric functions.

Theorem 5.6.3: *The collection* $\mathcal{T}_0 \equiv \{\cos nx : n = 0, 1, 2, \ldots\} \cup \{\sin nx : n = 1, 2, \ldots\}$ *is a complete orthogonal system for* $L^2[0, 2\pi]$.

Proof: By Theorem 5.6.1 for each $f \in \mathcal{C}[0, 2\pi]$ and $\epsilon > 0$, there exists a finite linear combination $\mathcal{D}_m(f, \cdot)$ of $\{\cos nx : n = 0, 1, 2, \ldots\}$ and $\{\sin nx : n = 1, 2, \ldots\}$ such that

$$\begin{aligned} &\int_0^{2\pi} |f - \mathcal{D}_m(f, \cdot)|^2 dx \\ &\quad \leq 2\pi (\sup\{|f(x) - \mathcal{D}_m(f, x)| : x \in [0, 2\pi]\})^2 < \epsilon^2. \end{aligned}$$

Also, from Theorem 2.3.14 it is known that given any $g \in L^2[0, 2\pi]$, and any $\epsilon > 0$, there is a $f \in \mathcal{C}[0, 2\pi]$ such that $\int_0^{2\pi} |f - g|^2 dx < \epsilon^2$. Thus, for any $g \in L^2[0, 2\pi]$, $\epsilon > 0$, there is a $f \in \mathcal{C}[0, 2\pi]$ and an $m \geq 1$ such that

$$\|g - \mathcal{D}_m(f, \cdot)\|_2 < 2\epsilon.$$

That is, the set $\mathcal{T}$ of all finite linear combinations of the functions in the class $\mathcal{T}_0$ is dense in $L^2[0, 2\pi]$. Further, it is easy to verify that $h_1, h_2 \in \mathcal{T}_0$, $h_1 \neq h_2$ implies

$$\int_0^{2\pi} h_1(x) h_2(x) dx = 0,$$

i.e., $\mathcal{T}_0$ is an orthogonal family. Since $\mathcal{T}_0$ is orthogonal and $\mathcal{T}$ is dense in $L^2[0, 2\pi]$, $\mathcal{T}_0$ is complete. $\square$

Definition 5.6.2: A function in $\mathcal{T}$ is called a *trigonometric polynomial.*

Thus, the above theorem says that trigonometric polynomials are dense in $L^2[0, 2\pi]$. Completeness of $\mathcal{T}$ in $L^2[0, 2\pi]$ and the results of Section 3.3 lead to

Theorem 5.6.4: *Let $f \in L^2[0, 2\pi]$. Let $\{(a_n, b_n), n = 0, 1, 2, \ldots\}$ and $\{s_n(f, \cdot)\}_{n \geq 0}$ be the associated Fourier coefficient sequences and partial sum sequence of the Fourier series for f as in Definition 5.6.1. Then*

(i) $s_n(f, \cdot) \to f$ *in* $L^2[0, 2\pi]$,

(ii) $\sum_{n=0}^{\infty}(\breve{a}_n^2 + \breve{b}_n^2) = \frac{1}{2\pi}\int_0^{2\pi} |f|^2 dx$ *where for* $n \geq 0$, $\breve{a}_n = a_n/c_n$, *with* $c_n^2 = \frac{1}{2\pi}\int_0^{2\pi}(\cos nx)^2 dx = \frac{1}{2}$, *and for* $n \geq 1$, $\breve{b}_n = \frac{b_n}{d_n}$ *with* $d_n^2 = \frac{1}{2\pi}\int_0^{2\pi}(\sin nx)^2 dx = \frac{1}{2}$.

(iii) Further, if f, $g \in L^2[0, 2\pi]$, *then* $\frac{1}{2\pi}\int_0^{2\pi} fg\, dx = a_0\alpha_0 + \sum_{n=1}^{\infty}\left(\frac{a_n\alpha_n}{c_n^2} + \frac{b_n\beta_n}{d_n^2}\right)$, *where* $\{(a_n, b_n) : n = 0, 1, 2, \ldots\}$ *and* $\{(\alpha_n, \beta_n) : n = 0, 1, 2, \ldots\}$ *are, respectively, the Fourier coefficients of* f *and* g.

Clearly (ii) above is a restatement of Bessel's equality. Assertion (iii) is known as the *Parseval identity.* As for convergence pointwise or almost everywhere, A. N. Kolmogorov showed in 1926 (see Körner (1989)) that there exists an $f \in L^1[0, 2\pi]$ such that $\overline{\lim}_{n\to\infty}|s_n(f, x)| = \infty$ everywhere on $[0, 2\pi]$. This led to the belief that for $f \notin \mathcal{C}[0, 2\pi]$, the mean square convergence of (i) in Theorem 5.6.4 cannot be improved upon. But L. Carleson showed in 1964 (see Körner (1989)) that for f in $L^2[0, 2\pi]$, $s_n(f, \cdot) \to f(\cdot)$ almost everywhere. Finally, turning to $L^1[0, 2\pi]$, one has the following:

Theorem 5.6.5: *Let $f \in L^1[0, 2\pi]$. Let $\{(a_n, b_n) : n \geq 0\}$ be as in (6.1) and satisfy $\sum_{n=0}^{\infty}(|a_n| + |b_n|) < \infty$. Let $s_n(f, \cdot)$ be as in (6.2). Then $s_n(f, \cdot)$ converges uniformly on $[0, 2\pi]$ and the limit coincides with f almost everywhere.*

Proof: Note that $\sum_{n=0}^{\infty}(|a_n| + |b_n|) < \infty$ implies that the sequence $\{s_n(f, \cdot)\}_{n \geq 0}$ is a Cauchy sequence in the Banach space $\mathcal{C}[0, 2\pi]$ with the sup-norm. Thus, there exists a g in $\mathcal{C}[0, 2\pi]$ such that $s_n(f, \cdot) \to g$ uniformly on $[0, 2\pi]$. It is easy to check that this implies that g and f have the same Fourier coefficients. Set $h = g - f$. Then $h \in L^1[0, 2\pi]$ and the Fourier coefficients of h are all zero. This implies that h is orthogonal to the members of the class $\mathcal{T}$, which in turn yields that h is orthogonal to all continuous functions in $\mathcal{C}[0,2\pi]$, i.e.,

$$\int_0^{2\pi} h(x)k(x)dx = 0$$

for all $k \in \mathcal{C}[0,2\pi]$. Since $h \in L^1[0,2\pi]$ and for any interval $A \subset [0,2\pi]$, there exists a sequence $\{k_n\}_{n\geq 1}$ of uniformly bounded continuous functions, such that $k_n \to I_A$ a.e. (m), by the DCT,

$$\int h(x)I_A(x)dx = \lim_{n\to\infty} \int h(x)k_n(x)dx = 0.$$

This in turn implies that $h = 0$ a.e., i.e., $g = f$ a.e. $\square$

Remark 5.6.1: If $f \in L^2[0,2\pi]$, the Fourier coefficients $\{a_n, b_n\}$ are square summable and hence go to zero as $n \to \infty$. What if $f \in L^1[0,2\pi]$?

If $f \in L^1[0,2\pi]$, one can assert the following:

Theorem 5.6.6: (*Riemann-Lebesgue lemma*). *Let* $f \in L^1[0,2\pi]$. *Then*

$$\lim_{n\to\infty} \int_0^{2\pi} f(x)\cos nx\, dx = 0 = \lim_{n\to\infty} \int_0^{2\pi} f(x)\sin nx\, dx.$$

Proof: The lemma holds if $f = I_A$ for any interval $A \subset [0,2\pi]$ and since step functions (i.e., linear combinations of indicator functions of intervals) are dense in $L^1[0,2\pi]$, the lemma is proved. $\square$

It can be shown that the mapping $f \to \{(a_n, b_n)\}_{n\geq 0}$ from $L^1[0,2\pi]$ to bivariate sequences that go to zero as $n \to \infty$ is one-to-one but not onto (Rudin (1987), Chapter 5).

Remark 5.6.2: (*The complex case*). Let $T \equiv \{z : z = e^{\iota\theta}, 0 \leq \theta \leq 2\pi\}$ be the unit circle in the complex plane $\mathbb{C}$. Every function $g : T \to \mathbb{C}$ can be identified with a function f on $\mathbb{R}$ by $f(t) = g(e^{\iota t})$. Clearly, $f(\cdot)$ is periodic on $\mathbb{R}$ with period 2π. In the rest of this section, for $0 < p < \infty$, $L^p(T)$ will stand for the collection of all Borel measurable functions $f : [0,2\pi]$ to $\mathbb{C}$ such that $\int_{[0,2\pi]} |f|^p dm < \infty$ where $m(\cdot)$ is the Lebesgue measure. A *trigonometric polynomial* is a function of form

$$f(\cdot) \equiv \sum_{n=-k}^{k} \alpha_n e^{\iota nx} \equiv a_0 + \sum_{n=1}^{k} (a_n \cos nx + b_n \sin nx),$$

$k < \infty$, where $\{\alpha_n\}$, $\{a_n\}_{n\geq 0}$, and $\{b_n\}_{n\geq 0}$ are sequences of complex numbers.

The completeness of the trigonometric polynomials proved in Theorem 5.6.3 implies that the family $\{e^{\iota nx} : n = 0, \pm 1, \pm 2, \ldots\}$ is a complete orthonormal basis for $L^2(T)$, which is a complex Hilbert space.

Thus Theorem 5.6.4 carries over to this case.

Theorem 5.6.7:

(i) Let $f \in L^2(T)$. Then,

$$\sum_{n=-k}^{k} \alpha_n e^{\iota nx} \to f \quad in \quad L^2(T)$$

where

$$\alpha_n \equiv \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-\iota ns}dx, \; n \in \mathbb{Z}.$$

Further,

$$\sum_{n=-\infty}^{\infty} |\alpha_n|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f|^2 dm.$$

(ii) For any sequence $\{\alpha_n\}_{n\in\mathbb{Z}}$ of complex numbers such that $\sum_{n=-\infty}^{\infty} |\alpha_n|^2 < \infty$, the sequence $\left\{f_k(x) \equiv \sum_{n=-k}^{k} \alpha_n e^{\iota nx}\right\}_{k\geq 1}$ converges in $L^2(T)$ to a unique f such that

$$\alpha_n = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-\iota nx}dx.$$

(iii) For any $f,\ g \in L^2(T)$,

$$\sum_{n=-\infty}^{\infty} \alpha_n \bar{\beta}_n = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}dx$$

where

$$\begin{aligned} \alpha_n &= \hat{f}(n) = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-\iota nx}dx, \\ \beta_n &= \hat{g}(n) = \frac{1}{2\pi}\int_0^{2\pi} g(x)e^{-\iota nx}dx, \; n \in \mathbb{Z}. \end{aligned}$$

Further,

$$\sum_{n=-\infty}^{\infty} |\alpha_n \beta_n| < \infty.$$

(iv) $L^2(T)$ is isomorphic to $\ell_2(\mathbb{Z})$, the Hilbert space of all square summable sequences of complex numbers on $\mathbb{Z}$.

Similarly, Theorem 5.6.5 carries over to the complex case.

Theorem 5.6.8: *Let $f \in L^1(T)$. Suppose*

$$\sum_{n=-\infty}^{\infty} |\hat{f}(n)| < \infty$$

where

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-\iota nx} dx, \ n \in \mathbb{Z}.$$

Then

$$s_n(f, x) \equiv \sum_{j=-n}^{n} \hat{f}(j) e^{-\iota jx}$$

converges uniformly on $[0, 2\pi]$ and the limit coincides with f a.e. and hence f is continuous a.e.

5.7 Fourier transforms on $\mathbb{R}$

In this section and in Section 5.8, let $L^p(\mathbb{R})$ stand for

$$L^p(\mathbb{R}) \equiv \{f : f : \mathbb{R} \to \mathbb{C}, \text{ Borel measurable, } \int_{\mathbb{R}} |f|^p dm < \infty\} \tag{7.1}$$

where $m(\cdot)$ is Lebesgue measure. Also, for $f \in L^1(\mathbb{R})$, $\int_{\mathbb{R}} f dm$ will often be written as $\int f(x)dx$. Let

$$\mathcal{C}_0 \equiv \{f : f : \mathbb{R} \to \mathbb{C}, \text{ continuous and } \lim_{|x| \to \infty} f(x) = 0\}. \tag{7.2}$$

Definition 5.7.1: For $f \in L^1(\mathbb{R})$, $t \in \mathbb{R}$,

$$\hat{f}(t) \equiv \int f(x) e^{-\iota tx} dx \tag{7.3}$$

is called the *Fourier transform of f*.

Proposition 5.7.1: *Let $f \in L^1(\mathbb{R})$ and $\hat{f}(\cdot)$ be as in (7.3). Then*

(i) $\hat{f}(\cdot) \in \mathcal{C}_0$.

(ii) If $f_a(x) \equiv f(x - a)$, $a \in \mathbb{R}$, then $\hat{f}_a(t) = e^{\iota ta} \hat{f}(t)$, $t \in \mathbb{R}$.

Proof:

(i) For any $t \in \mathbb{R}$, $t_n \to t \Rightarrow e^{\iota t_n x} f(x) \to e^{\iota tx} f(x)$ for all $x \in \mathbb{R}$ and since $|e^{\iota t_n x} f(x)| \leq |f(x)|$ for all n and x, by the DCT $\hat{f}(t_n) \to \hat{f}(t)$. To show that $\hat{f}(t) \to 0$ as $|t| \to \infty$, the same proof as that of Theorem 5.6.6 works. Thus, it holds if $f = I_{[a,b]}$, for a, $b \in \mathbb{R}$, $a < b$ and since the step functions are dense in $L^1(\mathbb{R})$, it holds for all $f \in L^1(\mathbb{R})$.

(ii) This is a consequence of the translation invariance of $m(\cdot)$, i.e., $m(A+a) = m(A)$ for all $A \in \mathcal{B}(\mathbb{R})$, $a \in \mathbb{R}$. □

The continuity of $\hat{f}(\cdot)$ can be strengthened to differentiability if, in addition to $f \in L^1(\mathbb{R})$, $xf(x) \in L^1(\mathbb{R})$. More generally, if $f \in L^1(\mathbb{R})$ and $x^k f(x) \in L^1(\mathbb{R})$ for some $k \geq 1$, then $\hat{f}(\cdot)$ is differentiable k-times with all derivatives $\hat{f}^{(r)}(t) \to 0$ as $|t| \to \infty$ for $r \leq k$ (Problem 5.22).

Proposition 5.7.2: *Let f, $g \in L^1(\mathbb{R})$ and $f * g$ be their convolution as defined in (4.7). Then*

$$\widehat{f * g} = \hat{f}\hat{g}. \tag{7.4}$$

Proof:

$$\begin{aligned} \widehat{f * g}(t) &= \int_{\mathbb{R}} e^{-\iota tx} \Big(\int_{\mathbb{R}} f(x-u) g(u) du \Big) dx \\ &= \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} e^{-\iota t(x-u)} f(x-u) e^{-\iota tu} g(u) du \Big) dx \\ &= \int_{\mathbb{R}} (f_t * g_t)(x) dx \end{aligned} \tag{7.5}$$

where $f_t(x) = e^{-\iota tx} f(x)$, $g_t(x) = e^{-\iota tx} g(x)$. Thus, by Proposition 5.4.4

$$\begin{aligned} \widehat{f * g}(t) &= \Big(\int_{\mathbb{R}} f_t(x) dx \Big) \Big(\int_{\mathbb{R}} g_t(x) dx \Big) \\ &= \hat{f}(t) \hat{g}(t). \end{aligned}$$

□

The process of recovering f from $\hat{f}$ (i.e., that of finding an inversion formula) can be developed along the lines of Fejer's theorem (Theorem 5.6.1).

Theorem 5.7.3: (*Fejer's theorem*). *Let $f \in L^1(\mathbb{R})$, $\hat{f}(\cdot)$ be as in (7.3) and*

$$S_T(f, x) \equiv \frac{1}{2\pi} \int_{-T}^{T} \hat{f}(t) e^{\iota tx} dt, \ T \geq 0, \tag{7.6}$$

$$D_R(f, x) \equiv \frac{1}{R} \int_0^R S_T(f, x) dT, \ R \geq 0. \tag{7.7}$$

(i) If f is continuous at x_0 and f is bounded on $\mathbb{R}$, then

$$\lim_{R \to \infty} D_R(f, x_0) = f(x_0). \tag{7.8}$$

(ii) If f is uniformly continuous and bounded on $\mathbb{R}$, then

$$\lim_{R \to \infty} D_R(f, \cdot) = f(\cdot) \text{ uniformly on } \mathbb{R}. \tag{7.9}$$

(iii) As $R \to \infty$,

$$D_R(f, \cdot) \to f(\cdot) \quad \textit{in} \quad L^1(\mathbb{R}). \tag{7.10}$$

(iv) If $f \in L^p(\mathbb{R})$, $1 \le p < \infty$, then as $R \to \infty$,

$$D_R(f, \cdot) \to f(\cdot) \quad \textit{in} \quad L^p(\mathbb{R}). \tag{7.11}$$

Corollary 5.7.4: (*Uniqueness theorem*). *If f and $g \in L^1(\mathbb{R})$ and $\hat{f}(\cdot) = \hat{g}(\cdot)$, then $f = g$ a.e. (m).*

Proof: Let $h = f - g$. Then $h \in L^1(\mathbb{R})$ and $\hat{h}(\cdot) \equiv 0$. Thus, $S_T(h, \cdot) \equiv 0$ and $D_R(h, \cdot) \equiv 0$ where $S_T(h, \cdot)$ and $D_R(h, \cdot)$ are as in (7.6) and (7.7). Hence by Theorem 5.7.3 (iii), $h = 0$ a.e. (m), i.e., $f = g$ a.e. (m). $\square$

Corollary 5.7.5: (*Inversion formula*). *Let $f \in L^1(\mathbb{R})$ and $\hat{f} \in L^1(\mathbb{R})$. Then*

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t) e^{\iota t x} dx \quad \textit{a.e.} \quad (m). \tag{7.12}$$

Proof: Since $\hat{f} \in L^1(\mathbb{R})$, by the DCT,

$$S_T(f, x) \to \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(t) e^{\iota t x} dt \quad \text{as} \quad T \to \infty$$

for all x in $\mathbb{R}$ and hence $D_R(f, x)$ has the same limit as $R \to \infty$. Now (7.12) follows from (7.10). $\square$

The following results, i.e., Lemma 5.7.6 and Lemma 5.7.7, are needed for the proof of Theorem 5.7.3. The first one is an analog of Lemma 5.6.2.

Lemma 5.7.6: (*Fejer*). *For $R > 0$, let*

$$K_R(x) \equiv \frac{1}{2\pi} \frac{1}{R} \int_0^R \Big(\int_{-T}^{T} e^{\iota t x} dt \Big) dT. \tag{7.13}$$

Then

(i)

$$K_R(x) = \begin{cases} \frac{1}{\pi} \frac{(1 - \cos Rx)}{x^2}, & x \neq 0 \\ \frac{R}{2\pi} & x = 0, \end{cases} \tag{7.14}$$

and hence $K_R(\cdot) \ge 0$.

(ii) For $\delta > 0$,

$$\int_{|x| \ge \delta} K_R(x) dx \to 0 \quad \textit{as} \quad R \to \infty. \tag{7.15}$$

(iii)

$$\int_{-\infty}^{\infty} K_R(x)dx = 1. \tag{7.16}$$

Proof:

(i) $K_R(0) = \frac{1}{2\pi R}\int_0^R (2T)dT = \frac{1}{2\pi}R$. For $x \neq 0$ and $R > 0$,

$$\begin{aligned} K_R(x) &= \frac{1}{2\pi R}\int_0^R \frac{2\sin Tx}{x}dT \\ &= \frac{1}{2\pi R}\,\frac{2(1-\cos Rx)}{x^2}. \end{aligned}$$

(ii) For $\delta > 0$,

$$\begin{aligned} 0 \leq \int_{|x|\geq\delta} K_R(x)dx &\leq \frac{2}{2\pi R}\int_{|x|\geq\delta}\frac{1}{x^2}\,dx \\ &= \frac{2}{\pi R}\,\frac{1}{\delta} \to 0 \quad \text{as} \quad R \to \infty. \end{aligned}$$

(iii)

$$\begin{aligned} \int_{-\infty}^{\infty} K_R(x)dx &= \frac{2}{\pi}\,\frac{1}{R}\int_0^{\infty}\Big(\frac{1-\cos Rx}{x^2}\Big)dx \\ &= \frac{2}{\pi}\int_0^{\infty}\Big(\frac{1-\cos u}{u^2}\Big)du. \end{aligned}$$

Now

$$\begin{aligned} &\int_0^{\infty}\frac{1-\cos u}{u^2}\,du \\ &= \lim_{L\to\infty}\int_0^L\Big(\frac{1-\cos u}{u^2}\Big)du \quad \text{(by the MCT)} \\ &= \lim_{L\to\infty}\int_0^L\Big(\int_0^u \sin x\,dx\Big)\frac{1}{u^2}\,du \\ &= \lim_{L\to\infty}\int_0^L \sin x\Big(\int_x^L \frac{1}{u^2}\,du\Big)dx \quad \text{(by Fubini's theorem)} \\ &= \lim_{L\to\infty}\Big(\int_0^L \frac{\sin x}{x}\,dx - \frac{1}{L}\int_0^L \sin x\,dx\Big) \\ &= \lim_{L\to\infty}\int_0^L \frac{\sin x}{x}\,dx \quad \text{since} \quad \Big|\int_0^L \sin x\,dx\Big| \leq 1. \end{aligned}$$

Thus,

$$\int_0^\infty \frac{1-\cos u}{u^2}du = \int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2} \tag{7.17}$$

(cf. Problem 5.9). Hence (iii) follows. □

Lemma 5.7.7: *Let $f \in L^p(\mathbb{R})$, $0 < p < \infty$. Then*

$$\int_{-\infty}^\infty |f(x-u) - f(x)|^p dx \to 0 \quad as \quad |u| \to 0. \tag{7.18}$$

Proof: The lemma holds if $f \in \mathcal{C}_K$, i.e., if f is continuous on $\mathbb{R}$ (with values in $\mathbb{C}$) and vanishes outside a bounded interval. By Theorem 2.3.14, such functions are dense in $L^p(\mathbb{R})$. So given $f \in L^p(\mathbb{R})$, $0 < p < \infty$ and $\epsilon > 0$, let $g \in \mathcal{C}_K$ be such that

$$\int |f-g|^p dm < \epsilon.$$

For any $0 < p < \infty$, there is a $0 < c_p < \infty$ such that for all $x, y, z \in (0, \infty)$, $|x+y+z|^p \le c_p(|x|^p + |y|^p + |z|^p)$. Then,

$$\begin{aligned}
&\int |f(x-u) - f(x)|^p dx \\
&\le \quad c_p \int \Big(|f(x-u) - g(x-u)|^p + |g(x-u) - g(x)|^p \\
&\qquad + \int |g(x) - f(x)|^p\Big) dx \\
&= \quad c_p\Big(2\epsilon + \int |g(x-u) - g(x)|^p\Big) du.
\end{aligned}$$

So

$$\varlimsup_{|u|\to 0} \int |f(x-u) - f(x)|^p dx \le c_p\, 2\epsilon.$$

Since $\epsilon > 0$ is arbitrary, the lemma is proved. □

Proof of Theorem 5.7.3: From (7.7)

$$\begin{aligned}
D_R(f,x) &\equiv \frac{1}{2\pi R}\int_0^R \Big(\int_{-T}^T e^{\iota t x}\Big(\int_{-\infty}^\infty e^{-\iota t y} f(y)dy\Big)dt\Big)dT \\
&= \frac{1}{2\pi}\frac{1}{R}\int_0^R \Big(\int_{-T}^T \Big(\int_{-\infty}^\infty e^{\iota t u} f(x-u)du\Big)dt\Big)dT.
\end{aligned}$$

Now Fubini's theorem yields

$$\begin{aligned}
D_R(f,x) &= \int_{-\infty}^\infty f(x-u)\Big(\frac{1}{2\pi}\frac{1}{R}\int_0^R \Big(\int_{-T}^T e^{\iota t u}dt\Big)dR\Big)du \\
&= \int_{-\infty}^\infty f(x-u)K_R(u)du \qquad (7.19)
\end{aligned}$$

where $K_R(\cdot)$ is as in (7.13).

Now let f be continuous at x_0 and bounded on $\mathbb{R}$ by M_f. Fix $\epsilon > 0$ and choose $\delta > 0$ such that $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \epsilon$. From (7.16) and (7.19),

$$D_R(f, x_0) - f(x_0) = \int \big(f(x_0 - u) - f(x_0)\big) K_R(u) du$$

implying

$$\begin{aligned} |D_R(f, x_0) - f(x_0)| \quad &\leq \quad \int_{|u|<\delta} |f(x_0 - u) - f(x_0)| K_R(u) du \\ &\qquad + \int_{|u|\geq\delta} |f(x_0 - u) - f(x_0)| K_R(u) du \\ &\quad < \epsilon + 2M_f \int_{|u|\geq\delta} K_R(u) du. \end{aligned}$$

Now from (7.15), it follows that

$$\overline{\lim_{R\to\infty}} |D_R(f, x_0) - f(x_0)| \leq \epsilon,$$

proving (i).

The proof of (ii) is similar to this and is omitted.

Clearly (iv) implies (iii). To establish (iv), note that for $1 \leq p < \infty$, by Jensen's inequality (which applies since $K_R(u) \geq 0$ and $\int K_R(u) du = 1$), for x in $\mathbb{R}$,

$$|D_R(f, x) - f(x)|^p \leq \int \big|\big(f(x - u) - f(x)\big)\big|^p K_R(u) du$$

and hence

$$\int |D_R(f, x) - f(x)|^p dx \leq \int \Big(\int |f(x - u) - f(x)|^p dx\Big) K_R(u) du.$$

Now (7.18) and the arguments in the proof of (i) yield (iv). □

5.8 Plancherel transform

If $f \in L^2(\mathbb{R})$, it need not be in $L^1(\mathbb{R})$ and so the Fourier transform is not defined. However, it is possible to extend the definition using an approximation procedure due to Plancherel.

Proposition 5.8.1: *Let $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and let $\hat{f}(t) \equiv \int f(x) e^{-\iota t x} dx$, $t \in \mathbb{R}$. Then, $\hat{f} \in L^2(\mathbb{R})$ and*

$$\int |f|^2 dm = \frac{1}{2\pi} \int |\hat{f}|^2 dm. \tag{8.1}$$

Proof: Let $\tilde{f}(x) = \overline{f(-x)}$ and $g = f * \tilde{f}$. Since f and $\tilde{f}$ are in $L^1(\mathbb{R})$, g is well defined and is in $L^1(\mathbb{R})$. Further by Cauchy-Schwarz inequality,

$$\begin{aligned} |g(x_1) - g(x_2)| &\leq \int |f(x_1 - u) - f(x_2 - u)|\,|\tilde{f}(u)|du \\ &\leq \Big(\int |f(x_1 - u) - f(x_2 - u)|^2 du\Big)^{1/2}\Big(\int |\tilde{f}(u)|^2 du\Big). \end{aligned}$$

By Lemma 5.7.7, the right side goes to zero uniformly as $|x_1 - x_2| \to 0$. Thus $g(\cdot)$ is uniformly continuous. Also, by Cauchy-Schwarz inequality,

$$\begin{aligned} |g(x)| &= \Big|\int f(x-u)\tilde{f}(u)du\Big| \\ &\leq \Big(\int |f(u)|^2 du\Big)^{1/2}\Big(\int |\tilde{f}(u)|^2 du\Big)^{1/2} \\ &= \int |f(u)|^2 du, \end{aligned}$$

and hence $g(\cdot)$ is bounded. Thus, g is continuous, bounded, and integrable on $\mathbb{R}$. By Theorem 5.7.3 (Fejer's theorem)

$$D_R(g, 0) \to g(0) \quad \text{as} \quad R \to \infty. \tag{8.2}$$

But

$$g(0) = \int f(u)\overline{f(-u)}du = \int |f|^2 dm. \tag{8.3}$$

By Proposition 5.7.2, $\hat{g}(t) = \hat{f}(t)\hat{\tilde{f}}(t)$. Also, note that

$$\begin{aligned} \hat{\tilde{f}}(t) &= \int \tilde{f}(x)e^{-\iota tx}dx \\ &= \int \overline{f(-x)}e^{-\iota tx}dx \\ &= \overline{\hat{f}(t)} \end{aligned}$$

and hence $\hat{g}(t) = |\hat{f}(t)|^2 \geq 0$. But

$$\begin{aligned} D_R(g, 0) &= \frac{1}{2\pi}\frac{1}{R}\int_0^R \Big(\int_{-T}^{T} \hat{g}(t)dt\Big)dT \\ &= \frac{1}{\pi}\int_0^R \hat{g}(t)\Big(1 - \frac{|t|}{R}\Big)dt. \end{aligned}$$

Since $\hat{g}(\cdot) \geq 0$, by the MCT,

$$\int_0^R \hat{g}(t)\Big(1 - \frac{|t|}{R}\Big)dt \uparrow \int_0^\infty \hat{g}(t)dt \quad \text{as} \quad R \uparrow \infty.$$

Thus, $\lim\limits_{R\to\infty} D_R(g,0) = \frac{1}{\pi}\int_0^\infty \hat{g}(t)dt$. Since $\hat{g}(-t) = \hat{g}(t)$ for all t in $\mathbb{R}$,

$$\lim_{R\to\infty} D_R(g,0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{g}(t)dt = \frac{1}{2\pi}\int |\hat{f}(t)|^2 dt. \tag{8.4}$$

Clearly, (8.2)–(8.4) imply (8.1). □

Proposition 5.8.2: *Let $f \in L^2(\mathbb{R})$ and $f_n(\cdot) \equiv f I_{[-n,n]}(\cdot)$, $n \geq 1$. Then, $\{f_n\}_{n\geq 1}$, $\{\hat{f}_n\}_{n\geq 1}$ are both Cauchy in $L^2(\mathbb{R})$ and hence, convergent in $L^2(\mathbb{R})$.*

Proof: Since for each $n \geq 1$, $f_n \in L^2(\mathbb{R}) \cap L^1(\mathbb{R})$, by Proposition 5.8.1,

$$\|f_{n_1} - f_{n_2}\|_2^2 = \frac{1}{2\pi}\int |\hat{f}_{n_1} - \hat{f}_{n_2}|^2 dm. \tag{8.5}$$

Since $f \in L^2(\mathbb{R})$, $f_n \to f$ in $L^2(\mathbb{R})$, $\{f_n\}_{n\geq 1}$ is Cauchy in $L^2(\mathbb{R})$. By (8.5) $\{\hat{f}_n\}_{n\geq 1}$ is also Cauchy in $L^2(\mathbb{R})$. □

Definition 5.8.1: Let $f \in L^2(\mathbb{R})$. The *Plancherel transform of* f, denoted by $\hat{f}$, is defined as $\lim_{n\to\infty} \hat{f}_n$, where

$$\hat{f}_n(t) = \int_{-n}^{n} e^{-\iota t x} f(x)dx \tag{8.6}$$

and the limit is taken in $L^2(\mathbb{R})$.

Theorem 5.8.3: *Let $f \in L^2(\mathbb{R})$ and $\hat{f}$ be its Plancherel transform. Then*

(i)

$$\int |f|^2 dm = \frac{1}{2\pi}\int |\hat{f}|^2 dm. \tag{8.7}$$

(ii) For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, the Plancherel transform coincides with the Fourier transform.

Proof: From Propositions 5.8.1 and 5.8.2 and the definition of $\hat{f}$,

$$\frac{1}{2\pi}\int |\hat{f}|^2 dm = \lim_{n\to\infty}\frac{1}{2\pi}\int |\hat{f}_n|^2 dm = \lim_{n\to\infty}\int |f_n|^2 dm = \int |f|^2 dm$$

proving (i). If $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\hat{f}_n$ defined in (8.6) converges as $n \to \infty$ pointwise to the Fourier transform of f (by the DCT). But by Definition 5.7.1, $\hat{f}_n$ converges in $L^2(\mathbb{R})$ to the Plancherel transform of f. So (ii) follows. □

It can also be shown that for f, $\hat{f}$ as in the above theorem, the following *inversion formula* holds:

$$\frac{1}{2\pi}\int_{-n}^{n} \hat{f}(t)e^{\iota t x}dt \to f(x) \quad \text{in} \quad L^2(\mathbb{R}) \tag{8.8}$$

and that the map $f \to \hat{f}$ is a Hilbert space isomorphism of $L^2(\mathbb{R})$ onto $L^2(\mathbb{R})$. For a proof, see Rudin (1987), Chapter 9. This in turn implies that if $f,\, g \in L^2(\mathbb{R})$ with Plancherel transforms $\hat{f}$ and $\hat{g}$, respectively, then

$$\int f\,\bar{g}\,dm = \frac{1}{2\pi}\int \hat{f}\,\bar{\hat{g}}\,dm. \tag{8.9}$$

This is known as the *Parseval identity.*

5.9 Problems

5.1 Verify that if μ_1 and μ_2 are finite measures on $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$, respectively, then $\mu_{12}(\cdot)$ and $\mu_{21}(\cdot)$, defined by (1.4) and (1.5), respectively, are measures on $(\Omega_1 \times \Omega_2,\, \mathcal{F}_1 \times \mathcal{F}_2)$.

(**Hint:** Use the MCT.)

5.2 Let $(\Omega, \mathcal{F}, \mu)$ be a complete measure space such that $\mathcal{F} \neq \mathcal{P}(\Omega)$ and for some $A_0 \in \mathcal{F}$ with $A_0 \neq \emptyset$, $\mu(A_0) = 0$. Let $B \in \mathcal{P}(\Omega) \setminus \mathcal{F}$. Then show that $(\mu \times \mu)(\Omega \times A_0) = 0$ but $B \times A_0 \notin \mathcal{F} \times \mathcal{F}$. Conclude that $(\Omega\times\Omega, \mathcal{F}\times\mathcal{F}, \mu\times\mu)$ is not complete. (An example of such a measure space $(\Omega, \mathcal{F}, \mu)$ is the space $([0,1], \mathcal{M}_L, m)$ where $\mathcal{M}_L$ is the Lebesgue σ-algebra on $[0,1]$ and m is the Lebesgue measure.)

5.3 Let μ_i, $i = 1, 2$ be two σ-finite measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $D_i = \{x : \mu_i(\{x\}) > 0\}, i = 1, 2$.

(a) Show that $D_1 \cup D_2$ is countable.

(b) Let $\phi_i(x) = \mu_i(\{x\})$ $x \in \mathbb{R}$, $i = 1, 2$. Show that ϕ_i is Borel measurable for $i = 1, 2$.

(c) Show that $\int \phi_1 d\mu_2 = \sum_{z\in D_1\cap D_2} \phi_1(z)\phi_2(z)$.

(d) Deduce (2.11) from (c).

5.4 Extend Theorem 5.2.3 as follows. Let F_1, F_2 be two nondecreasing right continuous functions on $[a, b]$. Then

$$\int_{(a,b]} F_1 dF_2 + \int_{(a,b]} F_2 dF_1 = F_1(b)F_2(b) - F_1(a)F_2(a) + \int_{(a,b]} \phi_1 d\mu_2,$$

where ϕ_1 is as in Problem 5.3.

5.5 Let $F_i : \mathbb{R} \to \mathbb{R}$ be nondecreasing and right continuous, $i = 1, 2$. Show that if $\lim_{b\uparrow\infty} F_1(b)F_2(b) = \lambda_1$ and $\lim_{a\downarrow-\infty} F_1(a)F_2(a) = \lambda_2$ exist and are finite, then

$$\int_{\mathbb{R}} F_1 dF_2 + \int_{\mathbb{R}} F_2 dF_1 = \lambda_1 - \lambda_2 + \int_{\mathbb{R}} \phi_1 d\mu_2$$

where ϕ_1 is as in Problem 5.3.

5.6 Let $(\Omega, \mathcal{F}, \mu)$ be a σ-finite measure space and f be a nonnegative measurable function. Then, $\int_\Omega f d\mu = \int_{[0,\infty)} \mu(\{f \geq t\})dt$.

(**Hint:** Consider the product space of $(\Omega, \mathcal{F}, \mu)$ with $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), m)$ and apply Tonelli's theorem to the function $g(\omega, t) = I(f(\omega) \geq t)$, after showing that g is $\mathcal{F} \times \mathcal{B}(\mathbb{R}_+))$-measurable.)

5.7 Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \to \mathbb{R}_+$ be a random variable.

(a) Show that for any $h : \mathbb{R}_+ \to \mathbb{R}_+$ that is absolutely continuous,

$$\begin{aligned} \int_\Omega h(X)dP &= h(0) + \int_{[0,\infty)} h'(t)P(X \geq t)dt \\ &= h(0) + \int_{(0,\infty)} h'(t)P(X > t)dt. \end{aligned}$$

(b) Show that for any $0 < p < \infty$,

$$\int_\Omega X^p dP = \int_{[0,\infty)} pt^{p-1}P(X \geq t)dt.$$

(c) Show that for any $0 < p < \infty$,

$$\int_\Omega X^{-p} dP = \frac{1}{\Gamma(p)} \int_{[0,\infty)} \psi_X(t)t^{p-1}dt,$$

where $\Gamma(p) = \int_{[0,\infty)} e^{-t}t^{p-1}dt, p > 0$, and $\psi_X(t) = \int_\Omega e^{-tX}dP$, $t \in \mathbb{R}_+$.

(**Hint:** (a) Apply Tonelli's theorem to the function $f(t, \omega) \equiv h'(t)I(X(w) \geq t)$ on the product measure space $([0,\infty) \times \Omega, \mathcal{B}([0,\infty)) \times \mathcal{F}, m \times P)$, where m is Lebesgue measure on $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$.)

5.8 Let $g : \mathbb{R}_+ \to \mathbb{R}_+$ and $f : \mathbb{R}^2 \to \mathbb{R}_+$ be Borel measurable. Let $A = \{(x, y) : x \geq 0, 0 \leq y \leq g(x)\}$.

(a) Show that $A \in \mathcal{B}(\mathbb{R}^2)$.

(b) Show that

$$\int_{\mathbb{R}_+} \Big(\int_{[0,g(x)]} f(x, y)m(dy) \Big) m(dx) = \int f I_A dm^{(2)},$$

where $m^{(2)}$ is Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$ and $m(\cdot)$ is Lebesgue measure on $\mathbb{R}$.

(c) If g is continuous and strictly increasing show that the two integrals in (b) equal

$$\int_{\mathbb{R}_+} \Big(\int_{[g^{-1}(y),\infty)} f(x,y) m(dx) \Big) m(dy).$$

5.9 (a) For $1 < A < \infty$, let

$$h_A(t) = \int_0^A e^{-xt} \sin x \, dx, \ \ t \geq 0.$$

Use integration by parts to show that

$$|h_A(t)| \leq \frac{1}{1+t^2} + e^{-t}$$

and

$$h_A(t) \to \frac{1}{1+t^2} \quad \text{as} \quad A \to \infty.$$

(b) Show using Fubini's theorem that for $0 < A < \infty$,

$$\int_0^\infty h_A(t) dt = \int_0^A \frac{\sin x}{x} \, dx.$$

(c) Conclude using the DCT that

$$\lim_{A\to\infty} \int_0^A \frac{\sin x}{x} \, dx = \int_0^\infty \frac{1}{1+t^2} \, dt.$$

(d) Using Theorem 4.4.1 and the fact that $\phi(x) \equiv \tan x$ is a (1–1) strictly monotone map from $(0, \frac{\pi}{2})$ to $(0,\infty)$ having the inverse map $\psi(\cdot)$ with derivative $\psi'(t) \equiv \frac{1}{1+t^2}$, $0 < t < \infty$, conclude that

$$\int_0^\infty \frac{1}{1+t^2} dt = \frac{\pi}{2} \ .$$

5.10 Show that $I \equiv \int_0^\infty e^{-x^2/2} dx = \sqrt{\frac{\pi}{2}}$.

(**Hint:** By Tonelli's theorem, $I^2 = \int_0^\infty \int_0^\infty e^{-\frac{(x^2+y^2)}{2}} dx dy$. Now use the change of variables $x = r\cos\theta$, $y = r\sin\theta$, $0 < r < \infty$, $0 < \theta < \frac{\pi}{2}$.)

5.11 Let μ be a finite measure on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Let f, $g : \mathbb{R} \to \mathbb{R}_+$ be nondecreasing. Show that

$$\mu(\mathbb{R}) \int fg \, d\mu \geq \Big(\int f d\mu \Big) \Big(\int g d\mu \Big).$$

(**Hint:** Consider $h(x_1, x_2) = \big(f(x_1) - f(x_2)\big)\big(g(x_1) - g(x_2)\big)$ on $\mathbb{R}^2$ and integrate w.r.t. $\mu \times \mu$.)

5.12 Let μ and λ be σ-finite measures on $\big(\mathbb{R}, \mathcal{B}(\mathbb{R})\big)$. Recall that

$$\nu(A) \equiv (\mu * \lambda)(A) \equiv \int\int I_A(x+y)d\mu d\lambda, \ A \in \mathcal{B}(\mathbb{R}).$$

(a) Show that for any Borel measurable $f : \mathbb{R} \to \mathbb{R}_+$, $f(x+y)$ is $\langle \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R})\rangle$-measurable from $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and

$$\int f d\nu = \int\int f(x+y)d\mu d\lambda.$$

(b) Show that

$$\nu(A) = \int_{\mathbb{R}} \mu(A-t)\lambda(dt), \ A \in \mathcal{B}(\mathbb{R}).$$

(c) Suppose there exist countable sets B_λ, B_μ such that $\mu(B_\mu^c) = 0 = \lambda(B_\lambda^c)$. Show that there exists a countable set B_ν such that $\nu(B_\nu^c) = 0$.

(d) Suppose that $\mu(\{x\}) = 0$ for all x in $\mathbb{R}$. Show that $\nu(\{x\}) = 0$ for all x in $\mathbb{R}$.

(e) Suppose that $\mu \ll m$ with $\frac{d\mu}{dm} = h$. Show that $\nu \ll m$ and find $\frac{d\nu}{dm}$ in terms of h, μ and λ.

(f) Suppose that $\mu \ll m$ and $\lambda \ll m$. Show that

$$\frac{d\nu}{dm} = \frac{d\mu}{dm} * \frac{d\lambda}{dm}.$$

5.13 (*Convolution of cdfs*). Let F_i, $i = 1, 2$ be cdfs on $\mathbb{R}$. Recall that a cdf F on $\mathbb{R}$ is a function from $\mathbb{R} \to \mathbb{R}_+$ such that it is nondecreasing, right continuous with $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to \infty$.

(a) Show that $(F_1 * F_2)(x) \equiv \int_{\mathbb{R}} F_1(x-u)dF_2(u)$ is well defined and is a cdf on $\mathbb{R}$.

(b) Show also that $(F_1 * F_2)(\cdot) = (F_2 * F_1)(\cdot)$.

(c) Suppose $t \in \mathbb{R}$ is such that $\int e^{tx}dF_i(x) < \infty$ for $i = 1, 2$. Show that $\int e^{tx}d(F_1 * F_2)(x) = \Big(\int e^{tx}dF_1(x)\Big)\Big(\int e^{tx}dF_2(x)\Big)$.

5.14 Let f, $g \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

(a) Show that if f is continuous and bounded on $\mathbb{R}$, then so is $f * g$.

(b) Show that if f is differentiable with a bounded derivative on $\mathbb{R}$, then so is $f * g$.

(**Hint:** Use the DCT.)

5.15 Let $f \in L^1(\mathbb{R})$, $g \in L^p(\mathbb{R})$, $1 \le p \le \infty$.

(a) Show that if $1 \le p < \infty$, then for all x in $\mathbb{R}$

$$\Big| \int |f(u)g(x-u)|du \Big|^p \le \Big(\int |f|dm \Big)^{p-1} \Big(\int |g(x-u)|^p |f(u)|du \Big)$$

and hence that

$$(f * g)(x) \equiv \int f(u)g(x-u)du$$

is well defined a.e. (m) and

$$\|f * g\|_p \le \|f\|_1 \, \|g\|_p$$

with "=" holding iff either $f = 0$ a.e. or $g = 0$ a.e.

(b) Show that if $p = \infty$ then

$$\|f * g\|_\infty \le \|f\|_1 \, \|g\|_\infty$$

and "=" can hold for some nonzero f and g.

(**Hint for (a):** Use Jensen's inequality with probability measure $d\mu = \frac{|f|dm}{\|f\|_1}$ if $\|f\|_1 > 0$.)

5.16 Let $1 \le p \le \infty$ and $q = 1 - \frac{1}{p}$. Let $f \in L^p(\mathbb{R})$, $g \in L^q(\mathbb{R})$.

(a) Show that $f * g$ is well defined and uniformly continuous.

(b) Show that if $1 < p < \infty$,

$$\lim_{|x| \to \infty} (f * g)(x) = 0.$$

(**Hint:** For (a) use Hölder's inequality and Lemma 5.7.7. For (b) approximate g by simple functions.)

5.17 Let $g : \mathbb{R} \to \mathbb{R}$ be infinitely differentiable and be zero outside a bounded interval.

(a) Let $f : \mathbb{R} \to \mathbb{R}$ be Borel measurable and $\int_A |f|dm < \infty$ for all bounded intervals A in $\mathbb{R}$. Show that $f * g$ is well defined and infinitely differentiable.

(b) Show that for any $f \in L^1(\mathbb{R})$, there exist a sequence $\{g_n\}_{n \ge 1}$ of such functions such that $f * g_n \to f$ in $L^1(\mathbb{R})$.

5.18 For $f \in L^1(\mathbb{R})$, let $f_\sigma = f * \phi_\sigma$ where $\phi_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$, $0 < \sigma < \infty$, $x \in \mathbb{R}$.

(a) Show that f_σ is infinitely differentiable.

(b) Show that if f is continuous and zero outside a bounded interval, then f_σ converges to g uniformly on $\mathbb{R}$ as $\sigma \to 0$.

(c) Show that if $f \in L^p(\mathbb{R})$, $1 \le p < \infty$, then $f_\sigma \to f$ in $L^p(\mathbb{R})$ as $\sigma \to 0$.

5.19 Let $f \in L^p(\mathbb{R})$, $1 \le p < \infty$ and $h(x) = \int_{A+x} f(u)du$ where A is a bounded Borel set and

$$A + x \equiv \{y : y = a + x, a \in A\}.$$

(a) Show that $h = f * g$ for some g bounded and with bounded support.

(b) Show that $h(\cdot)$ is continuous and that $\lim_{|x|\to\infty} h(x) = 0$.

(**Hint:** For $1 < p < \infty$, use Hölder's inequality and show that $m\big((A+x_1) \triangle (A + x_2)\big) \to 0$ as $|x_1 - x_2| \to 0$.)

5.20 (a) Verify the claims in Examples 5.4.1 directly.

(b) Verify the same using generating functions.

5.21 Let f be a probability density on $\mathbb{R}$, i.e., f is nonnegative, Borel measurable and $\int f dm = 1$. Show that $|\hat{f}(t)| < 1$ for all $t \neq 0$.

(**Hint:** If $|\hat{f}(t_0)| = 1$ for some $t_0 \neq 0$, show that $\int \big(1 - \cos t_0(x - \theta)\big) f(x)dx = 0$ for some θ.)

5.22 (a) Let $f \in L^1(\mathbb{R})$ and $x^k f(x) \in L^1(\mathbb{R})$ for some $k \ge 1$. Show that $\hat{f}(\cdot)$ is k-times differentiable on $\mathbb{R}$ with all derivatives $\hat{f}^{(r)}(t) \to 0$ as $|t| \to \infty$ for $r \le k$.

(b) Let $f \in L^1(\mathbb{R})$. Suppose $\int |t\hat{f}(t)|dt < \infty$. Show that there exists a function $g : \mathbb{R} \to \mathbb{R}$ such that it is differentiable, $\lim_{|x|\to\infty}(|g(x)| + |g'(x)|) = 0$ and $g = f$ a.e. (m). Extend this to the case where $\int |t^k \hat{f}(t)|dt < \infty$ for some $k > 1$.

(**Hint:** Consider $g(x) = \frac{1}{2\pi} \int e^{-\iota tx} \hat{f}(t)dt$.)

5.23 Let $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $x \in \mathbb{R}$.

(a) Show that $\hat{f}(\cdot)$ is real valued, differentiable and satisfies the ordinary differential equation $\hat{f}'(t) + t\hat{f}(t) = 0$, $t \in \mathbb{R}$ and $\hat{f}(0) = 1$. Find $\hat{f}(t)$.

(b) For μ in $\mathbb{R}$, $\sigma > 0$, let

$$f_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma}\, e^{-\frac{1}{2}(\frac{x-u}{\sigma})^2}.$$

Find $\hat{f}_{\mu,\sigma}$ and verify that for any (μ_i, σ_i), $i = 1, 2$,

$$f_{\mu_1,\sigma_1} * f_{\mu_2,\sigma_2} = f_{\mu_1+\mu_2,\sigma_1^2+\sigma_2^2}.$$

(**Hint for (b):** Use Fourier transforms and uniqueness.)

5.24 (*Rate of convergence of Fourier series*). Consider the function $h(x) = \frac{\pi}{2} - |x|$ in $-\pi \le x \le \pi$.

(a) Find $\hat{h}(n) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-\iota nx} h(x)dx$, $n = 0, \pm 1, \pm 2$.

(b) Show that $\sum_{n=-\infty}^{\infty} |\hat{h}(n)| < \infty$.

(c) Show that $S_n(h, x) \equiv \sum_{j=-n}^{+n} \hat{h}(j)e^{\iota jx}$ converges to $h(x)$ uniformly on $[-\pi, \pi]$.

(d) Verify that

$$\sup\{|S_n(h,x) - h(x)| : -\pi \le x \le \pi\} \le \frac{2}{\pi}\frac{1}{(n-1)},\ n \ge 2$$

and

$$|S_n(h, 0) - h(0)| \ge \frac{2}{\pi}\frac{1}{(n+2)}\ .$$

(**Remark:** This example shows that the Fourier series of a function can converge very slowly such as in this example where the rate of decay is $\frac{1}{n}$.)

5.25 Using Fejer's theorem (Theorem 5.6.1) prove Wierstrass' theorem on uniform approximation of a continuous function on a bounded closed interval by a polynomial.

(**Hint:** Show that on bounded intervals a trigonometric polynomial can be approximated uniformly by a polynomial using the power series representation of sine and cosine functions (see Section A.3).

5.26 Evaluate

$$\lim_{n\to\infty}\int_{-n}^{n} \frac{\sin \lambda x}{x} e^{\iota tx} dx,\ 0 < \lambda < \infty,\ 0 < x < \infty.$$

(**Hint:** For $0 < \lambda < \infty$, $\frac{\sin \lambda x}{x} \in L^2(\mathbb{R})$ and it is the Fourier transform of $f(t) = I_{[-\lambda,\lambda]}(\cdot)$. Now apply Plancherel theorem. Alternatively use Fubini's theorem and the fact $\lim\limits_{n\to\infty}\int_{-n}^{n} \frac{\sin y}{y} dy$ exists in $\mathbb{R}$.)

5.27 Find an example of a function $f \in L^2(\mathbb{R}) \cap \left(L^1(\mathbb{R})\right)^c$ such that its Plancherel transform $\hat{f} \in L^1(\mathbb{R})$.

(**Hint:** Examine Problem 5.26.)